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# Supporting Information

## Studies on the Synthesis and Opioid Agonistic Activities of Mitragynine-Related Indole Alkaloids: Discovery of Opioid Agonists Structurally Different from Other Opioid Ligands

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## Experimental

**Preparation of 9-ethoxycorynantheidine (6).** To a stirred solution of **5** (32 mg, 0.083 mmol) in benzene (3 mL) was successively added aqueous 15% NaOH solution (73 μL, 0.249 mmol), tetrabutylammonium hydrogensulfate (8.4 mg, 0.025 mmol), and then EtI (20 μL, 0.166 mmol) at 0 °C under argon atmosphere and the mixture was stirred at the room temperature for 1 h. The reaction mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by $SiO_2$ column chromatography (*n*-hexane/AcOEt 7:3) to give **6** (27 mg, 80%) as an amorphous powder. UV (MeOH) $\lambda_{max}$ : 293, 227 nm. IR $\nu_{max}$ ($CHCl_3$) : 1702, 1628 $cm^{-1}$. $^1$H-NMR ($CDCl_3$) δ : 7.69 (1H, br.s, *Na*-H), 7.43 (1H, s, H-17), 6.96 (1H, dd, *J*=8.0, 8.0, H-11), 6.86 (1H, d, *J*=7.3, H-12), 6.43 (1H, d, *J*=7.5, H-10), 4.10 (1H, m, 9-O$\underline{CH_2}CH_3$), 3.75 and 3.70 (each 3H, each s, 17-$OCH_3$ and 22-$OCH_3$), 3.15 (1H, br.d, *J*=11.2, H-3), 3.14 (1H, m, H-6), 3.05 (1H, m, H-15), 3.00 (1H, m, H-21), 2.95 (1H, m, H-6), 2.93

(1H, dd, *J*=9.3, 3.7, H-5), 2.55 (2H, each m, H-5 and H-14), 2.45 (1H, m, H-21), 1.80 (1H, m, H-14), 1.75 (1H, m, H-19), 1.63 (1H, m, H-20), 1.41 (1H, m, H-19), 1.40 (3H, dd, *J*=7.1, 7.1, 9-OCH$_2$CH$_3$), 0.86 (3H, dd, *J*=7.4, 7.4, H-18). $^{13}$C-NMR ($CDCl_3$) δ : 169.2 (C-22), 160.5(C-17), 153.0 (C-9), 137.0 (C-13), 133.4 (C-2), 121.2 (C-11), 117.1 (C-7), 111.5 (C-16), 107.9 (C-8), 107.2 (C-12), 99.7 (C-10), 63.4 (9-OCH$_2$CH$_3$), 61.5 (17-OCH$_3$), 61.2 (C-3), 57.7 (C-21), 53.7 (C-5), 51.4 (22-OCH$_3$), 40.5 (C-20), 40.0 (C-15), 29.9 (C-14), 23.9 (C-6), 19.4 (C-19), 14.9 (9-OCH$_2$CH$_3$), 12.8 (C-18). EI-MS *m/z* (%) : 412 ($M^+$, 99), 397 (35), 381 (10), 283 (27), 228 (100). HR-FABMS : calcd. for $C_{24}H_{33}O_4N_2$ [$MH^+$] : 413.2440, found: 413.2432.

**Preparation of 9-isopropoxycorynantheidine (7).** To a stirred solution of **5** (30 mg, 0.078 mmol) in benzene (3 mL) was successively added aqueous 15% NaOH solution (70 μL, 0.234 mmol), tetrabutylammonium hydrogensulfate (8.4 mg, 0.025 mmol), and then *i*-PrI (20 μL, 0.234 mmol) at 0 °C under argon atmosphere and the mixture was stirred at the room temperature for 8 h. The reaction mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by $SiO_2$ column chromatography (*n*-hexane/AcOEt 7:3) to give **7** (22 mg, 67%) as an amorphous powder. UV (MeOH) $\lambda_{max}$ : 293, 225 nm. $^1$H-NMR ($CDCl_3$) δ : 7.64 (1H, br.s, *Na*-H), 7.35 (1H, s, H-17), 6.88 (1H, dd, *J*=8.0, 8.0, H-11), 6.80 (1H, d, *J*=8.1, H-12), 6.38 (1H, d, *J*=7.8, H-10), 4.54 (1H, m, 9-OCH$_2$(CH$_3$)$_2$), 3.65 and 3.63 (each 3H, each s, 17-OCH$_3$ and 22-OCH$_3$), 3.08 (1H, br.d, *J*=11.0, H-3), 3.05 (1H, m, H-15), 3.02 (1H, m, H-6), 2.95 (1H, m, H-6), 2.92 (1H, m, H-21), 2.87 (1H, dd, *J*=11.2, 5.4, H-5), 2.50 (1H, m, H-14), 2.46 (2H, each m, H-21 and H-5), 1.70 (1H, m, H-14), 1.68 (1H, m, H-19), 1.55 (1H, br.d, *J*=13.2, H-20), 1.12 (1H, m, H-19), 1.30 and 1.28 (each 3H, each d, *J*=6.1 and 5.9, 9-OCH$_2$(CH$_3$)$_2$), 0.79 (3H, dd, *J*=7.3, 7.3, H-18). $^{13}$C-NMR ($CDCl_3$) δ : 169.2 (C-22), 160.5 (C-17), 152.4 (C-9), 137.5 (C-13), 133.6 (C-2), 121.8 (C-11), 118.0 (C-7), 111.4 (C-16) , 108.0 (C-8), 103.7 (C-12), 102.3 (C-10), 69.8 (9-OCH$_2$(CH$_3$)$_2$), 61.5 (17-OCH$_3$), 61.2 (C-3), 57.7 (C-21), 53.7 (C-5), 51.3 (22-OCH$_3$), 40.1 (C-20), 39.9(C-15), 29.9 (C-14), 23.9 (C-6), 22.3 and 22.4 (9-OCH$_2$(CH$_3$)$_2$ ), 19.1 (C-19), 12.8 (C-18). EI-MS *m/z* (%) : 426 ($M^+$, 100), 411 (39), 383 (11), 297 (17), 242 (78). HR-FABMS : calcd. for $C_{25}H_{35}O_4N_2$ [$MH^+$] : 427.2564, found: 427.2597.

**Preparation of (8).** To a stirred solution of **5** (53 mg, 0.138 mmol) in $CH_2Cl_2$ (0.5 mL) was successively added aqueous 15% NaOH solution (110 μL, 0.414 mmol), Adogen 464 [methyltrialkyl($C_8$-$C_{10}$)ammonium chloride] (0.01 mL, 0.028 mmol), and



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then methoxymethylchloride (56 μL, 0.552 mmol) at 0 °C under argon atmosphere and the mixture was stirred at the room temperature for 1 h. The reaction mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by $SiO_2$ column chromatography (3% MeOH-$CHCl_3$) to give **8** (42 mg, 71%) as an amorphous powder. UV (MeOH) $\lambda_{max}$: 291, 223 nm. $^1$H-NMR ($CDCl_3$) δ : 7.73 (1H, br.s, *Na*-H), 7.43 (1H, s, H-17), 6.99~6.93 (2H, m, H-11 and H-12), 6.65 (1H, dd, *J*=7.1, 1.2, H-10), 5.48 (1H, d, *J*=6.3, 9-O$\underline{CH_2}$OCH$_3$), 5.27 (1H, d, *J*=6.3, 9-O$\underline{CH_2}$OCH$_3$), 3.78 and 3.72 (each 3H, each s, 17-OCH$_3$ and 22-OCH$_3$), 3.51 (3H, s, 9-OCH$_2$O$\underline{CH_3}$), 3.17 (1H, m, H-3), 3.15 (1H, m, H-6), 3.05 (1H, m, H-21), 3.03 (2H, m, H-6 and H-15), 2.93 (1H, dd, *J*=10.7, 5.1, H-5), 2.57 (1H, m, H-14), 2.53 (1H, m, H-5), 2.45 (1H, dd, *J*=10.5, 2.4, H-21), 1.79 (1H, br.d, *J*=12.0, H-14), 1.73 (1H, m, H-19), 1.63 (1H, m, H-20), 1.21 (1H, m, H-19), 0.86 (3H, dd, *J*=7.3, 7.3, H-18). $^{13}$C-NMR ($CDCl_3$) δ : 169.2 (C-22), 160.5 (C-17), 151.5 (C-9), 137.6 (C-13), 134.2 (C-2), 121.8 (C-11), 118.3 (C-7), 111.6 (C-16), 107.6 (C-8), 105.1 (C-12), 103.4 (C-10), 94.5 (9-$\underline{OCH_2}$OCH$_3$), 61.5 (17-OCH$_3$), 61.3 (C-3), 57.8 (C-21), 56.0 (9-OCH$_2$O$\underline{CH_3}$), 53.8 (C-5), 51.3 (22-OCH$_3$), 40.7 (C-20), 39.9 (C-15), 29.9 (C-14), 23.9 (C-6), 19.1 (C-19), 12.8 (C-18). EI-MS *m/z* (%) : 428 ($M^+$, 100), 413 (36), 383 (16), 299 (18), 244 (88). HR-FABMS : calcd. for $C_{24}H_{33}O_5N_2$ [$MH^+$] : 429.2389, found: 429.2350.

**Preparation of 7-methoxy-7*H*-mitragynine (13).** A solution of idosobenzene diacetate (24 mg, 0.078 mmol) in dry MeOH (1 mL) was added dropwise at 0 °C to a solution of mitragynine (**1**) (30 mg, 0.078 mmol) in dry MeOH (3 mL) under argon atmosphere. After stirring at room temperature for 1 h, the mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by $SiO_2$ column chromatography (*n*-hexane/AcOEt 7:3) to give **14** (31 mg, 96%) as an amorphous powder. UV (MeOH) $\lambda_{max}$: 320 (sh), 246 (sh), 221 nm. IR $\nu_{max}$ ($CHCl_3$) : 2930, 1698, 1600 cm$^{-1}$. $^1$H-NMR ($CDCl_3$) δ : 7.44 (1H, s, H-17), 7.32 (1H, dd, *J*=7.9, 7.9, H-11), 7.25 (1H, d, *J*=7.8, H-12), 6.70 (1H, d, *J*=8.3, H-10), 3.87 (3H, s, 9-OCH$_3$), 3.87 (3H, s, 17-OCH$_3$) , 3.70 (3H, s, 22-OCH$_3$), 3.05 (1H, m, H-3), 3.03 (1H, m, H-21), 3.00 (1H, m, H-15), 2.86 (3H, s, 7-OCH$_3$), 2.83 (1H, m, H-5), 2.80 (1H, m, H-14), 2.70 (1H, m, H-6), 2.60 (1H, m, H-5), 2.48 (1H, br-d, *J*=8.8, H-21), 1.87 (1H, m, H-14), 1.70 (1H, m, H-19), 1.60 (2H, m, H-6 and H-20), 1.25 (1H, m, H-19), 0.82 (3H, dd, *J*=7.4, 7.4, H-18). $^{13}$C-NMR ($CDCl_3$) δ : 184.1 (C-2), 169.3 (C-22), 160.7 (C-17), 156.6 (C-9), 155.8 (C-13), 131.0 (C-11), 122.7 (C-8), 114.4 (C-12), 111.3 (C-16), 108.8 (C-



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10), 86.7 (C-7), 61.9 (17-$OCH_3$), 61.8 (C-3), 58.1 (C-21), 55.6 (9-$OCH_3$), 51.9 (7-$OCH_3$), 51.3 (22-$OCH_3$), 50.2 (C-5), 40.6 (C-20), 39.3 (C-15), 35.1 (C-6), 26.0 (C-14), 19.0 (C-19), 12.8 (C-18). EI-MS *m/z* (%) : 428 ($M^+$, 3), 397 (100), 256 (10). HR-FABMS : calcd. for $C_{24}H_{33}O_5N_2$ [$MH^+$] : 429.2389, found: 429.2364.

**Preparation of 7-ethoxy-7*H*-mitragynine (14).** A solution of idosobenzene diacetate (24 mg, 0.078 mmol) in dry EtOH (1 mL) was added dropwise at 0 °C to a solution of mitragynine (**1**) (30 mg, 0.078 mmol) in dry MeOH (0.5 mL) under argon atmosphere. After stirring at room temperature for 1.5 h, the mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by $SiO_2$ column chromatography (*n*-hexane/AcOEt 1:1) to give **15** (18 mg, 53%) as an amorphous powder. UV (MeOH) $\lambda_{max}$: 320 (sh), 246 (sh), 221 nm. IR $\nu_{max}$ ($CHCl_3$) : 2930, 1698, 1600 $cm^{-1}$. $^1$H-NMR ($CDCl_3$) δ : 7.45 (1H, s, H-17), 7.33 (1H, dd, *J*=7.9, 7.9, H-11), 7.25 (1H, d, *J*=7.6, H-12), 6.70 (1H, d, *J*=8.3, H-10), 3.86 (3H, s, 9-$OCH_3$), 3.81 (3H, s, 17-$OCH_3$) , 3.70 (3H, s, 22-$OCH_3$), 3.1～2.8 (2H, m, 7-O$\underline{CH_2}CH_3$), 3.05 (2H, m, H-3 and H-5), 3.00 (1H, m, H-15), 2.98 (1H, m, H-21), 2.79 (1H, m, H-14), 2.70 (1H, br.d, *J*=13.9, H-6), 2.57 (1H, m, H-5), 2.47 (1H, dd, *J*=11.5, 2.9, H-21), 1.87 (1H, br.d, *J*=13.4, H-14), 1.72 (1H, m, H-19), 1.60 (1H, m, H-20), 1.55 (1H, m, H-6), 1.25 (1H, m, H-19), 1.10 (3H, dd, *J*=7.0, 7.0, 7-$OCH_2\underline{CH_3}$), 0.82 (3H, dd, *J*=7.3, 7.3, H-18). $^{13}$C-NMR ($CDCl_3$) δ [ppm] : 184.6 (C-2), 169.4 (C-22), 160.7 (C-17), 156.5 (C-9), 155.5 (C-13), 130.7 (C-11), 123.6 (C-8), 114.3 (C-12), 111.3 (C-16), 108.8 (C-10), 86.3 (C-7), 62.1 (17-$OCH_3$), 61.8 (C-3), 59.9 (7-O$\underline{CH_2}OCH_3$), 58.2 (C-21), 55.5 (9-$OCH_3$), 51.3 (22-$OCH_3$), 50.2 (C-5), 40.6 (C-20), 39.4 (C-15), 35.4 (C-6), 26.0 (C-14), 19.0 (C-19), 15.5 (7-$OCH_2O\underline{CH_3}$), 12.8 (C-18). EI-MS *m/z* (%) : 442 ($M^+$, 18), 397 (100), 365 (8). HR-FABMS : calcd. for $C_{25}H_{35}O_5N_2$ [$MH^+$] : 443.2552, found: 443.2546.

**Preparation of 4-quinolone derivative (15).** A mixture of **1** (30 mg, 0.075 mmol) and NaH (50% suspension, 15 mg, 0.38 mmol) in dry DMF (0.5 mL) was stirred at room temperature under air for 15 h. The reaction mixture was poured onto the chilled water and was extracted with $CHCl_3$ five times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by $SiO_2$ column chromatography (5% MeOH-$CHCl_3$) to give **16** (15 mg, 49%) as an amorphous powder. UV (MeOH) $\lambda_{max}$: 334, 323, 233, 201 nm. $^1$H-NMR ($CDCl_3$) δ : 9.69 (1H, s, *Na*-H), 7.67 (1H, d, *J*=8.6, H-12), 7.47 (1H, dd, *J*=8.2, 8.2, H-11), 7.44 (1H, s, H-17), 6.80 (1H, d, *J*=7.9, H-10), 4.26 (1H, d, *J*=12.2, H-5), 4.06 (3H, s, 9-$OCH_3$),

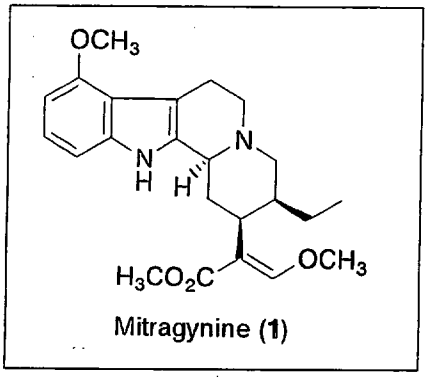
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3.74 (3H, s, 17-$OCH_3$) , 3.71 (3H, s, 22-$OCH_3$), 3.54 (1H, d, *J*=12.2, H-5), 3.37 (1H, d, *J*=11.3, H-3), 3.24 (1H, d, *J*=11.0, H-21), 3.02 (1H, d, *J*=12.7, H-15), 2.69 (1H, d, *J*=11.3, H-21), 2.54 (1H, m, H-14), 2.25 (1H, d, *J*=12.5, H-14), 1.73 (1H, m, H-19), 1.72 (1H, m, H-20), 1.25 (1H, m, H-19), 0.88 (3H, dd, *J*=7.3, 7.0, H-18). $^{13}$C-NMR ($CDCl_3$) δ : 168.3 (C-7), 169.3 (C-22), 160.7 (C-17), 156.2 (C-9), 155.5 (C-2), 151.3 (C-13), 128.1 (C-11), 122.9 (C-12), 115.3 (C-6), 111.0 (C-8), 103.9 (C-10), 69.4 (C-3), 61.6 (17-$OCH_3$), 56.4 (9-$OCH_3$), 53.5 (C-21), 52.9 (C-5), 51.3 (22-$OCH_3$), 40.5 (C-20), 39.8 (C-15), 27.6 (C-14), 19.5 (C-19), 12.9 (C-18). EI-MS *m/z* (%) : 412 ($M^+$, 100), 397 (28), 283 (21), 228 (50). HR-FABMS : calcd. for $C_{23}H_{29}O_5N_2$ [$MH^+$] : 413.2076, found: 413.2089.

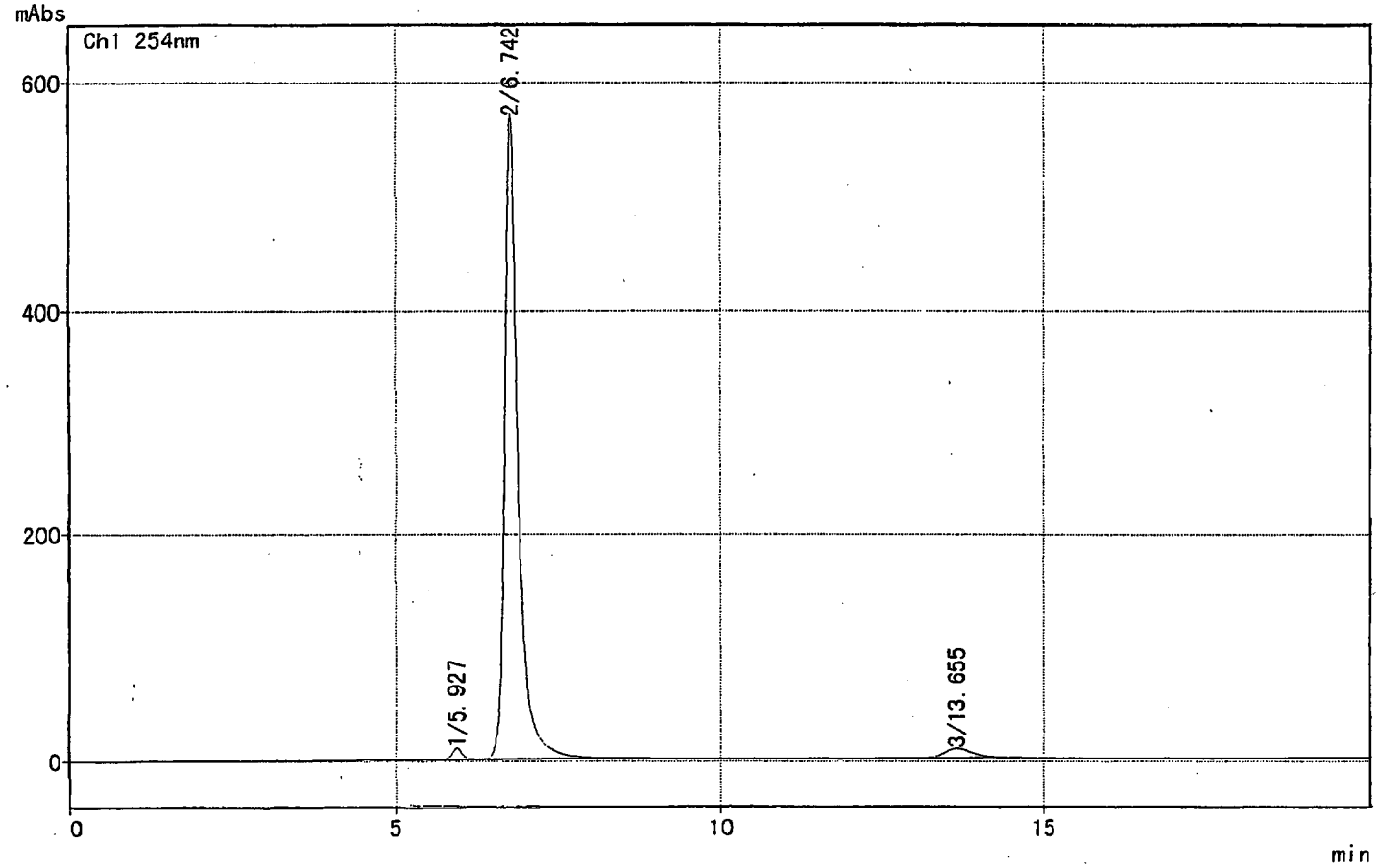
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## Condition

Column : CAPCELL PAK NH2 UG80
( SHISEIDO FINE CHEMICAL ) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 50% Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : Mitragynine



Mitragynine (**1**)




*** ピークレポート ***

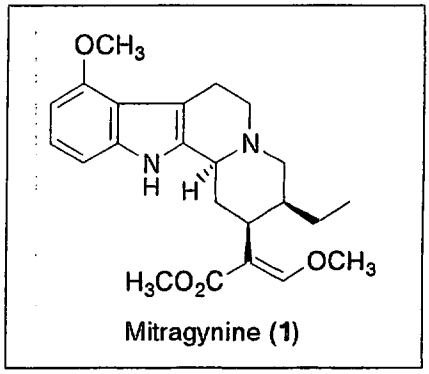
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| 2 | 1 | 8072496 | 6.742 |
| 3 | 1 | 252350 | 13.655 |
| | | 8435741 | |

Purity : 95.7 %

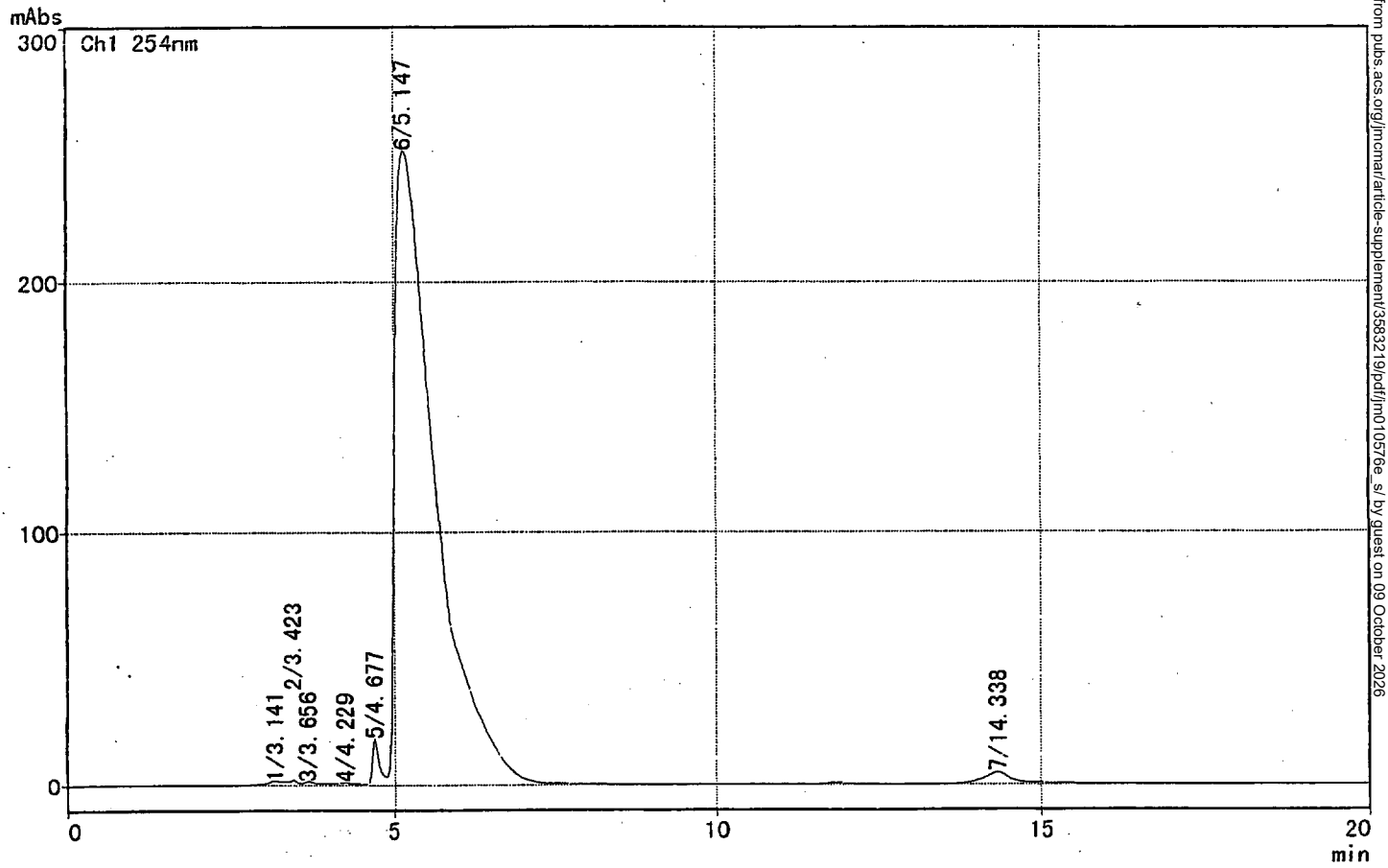
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## Condition

Column : YMC-Pack A-012 SIL
( YMC ) S-5μm
Column size : 6.0mm x 150mm
Flow Rate : 1.0mL/min
Eluent : 5% MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : Mitragynine



Mitragynine (**1**)




*** ピークレポート ***

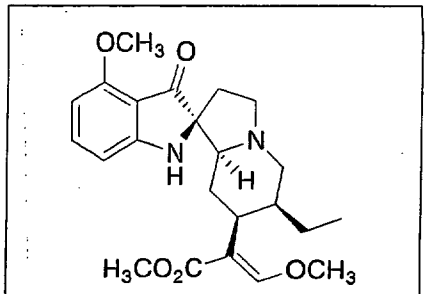
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| 3 | 1 | 5572 | 3.656 |
| 4 | 1 | 2474 | 4.229 |
| 5 | 1 | 135005 | 4.677 |
| 6 | 1 | 10916515 | 5.147 |
| 7 | 1 | 108241 | 14.338 |
| | | 11200057 | |

Purity : 97.5 %

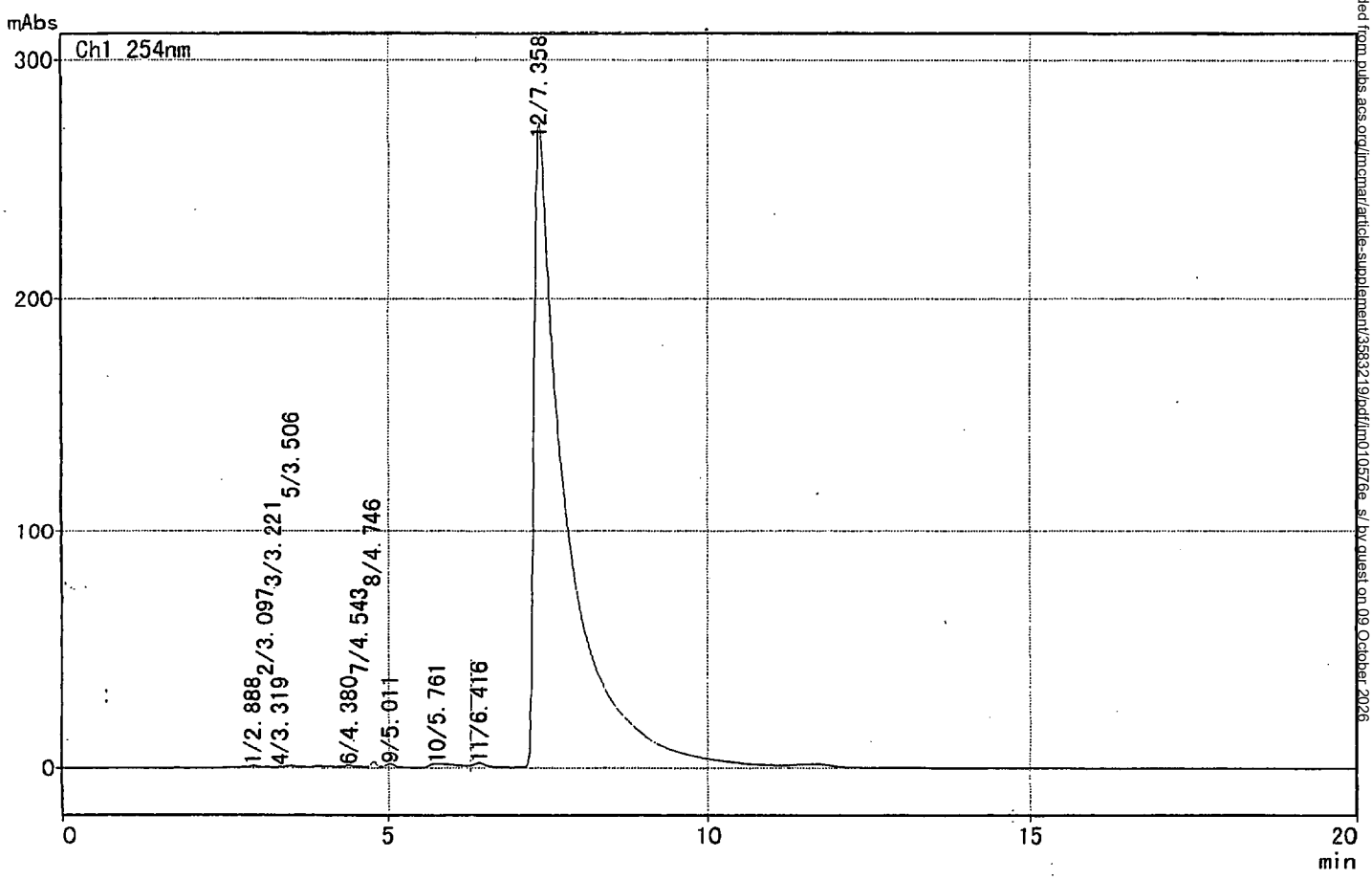
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# Condition

Column : PEGASIL Silica 60-5
( Senshu Pak )
Column size : 4.6mm x 250mm
Flow Rate : 1.0mL/min
Eluent : 3% MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : Mitragynine pseudoindoxyl




Mitragynine pseudoindoxyl (**2**)




*** ピークレポート ***

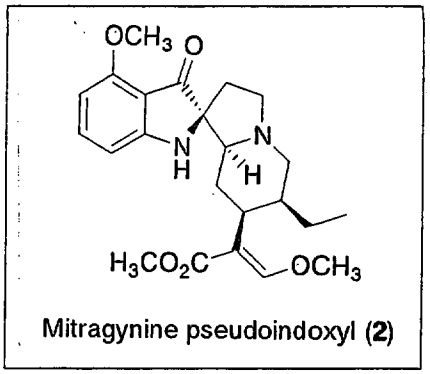
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| 2 | 1 | 2440 | 3.097 |
| 3 | 1 | 1309 | 3.221 |
| 4 | 1 | 2608 | 3.319 |
| 5 | 1 | 6443 | 3.506 |
| 6 | 1 | 3654 | 4.380 |
| 7 | 1 | 1490 | 4.543 |
| 8 | 1 | 13828 | 4.746 |
| 9 | 1 | 11755 | 5.011 |
| 10 | 1 | 40782 | 5.761 |
| 11 | 1 | 22444 | 6.416 |
| 12 | 1 | 9530087 | 7.358 |
| | | 9641805 | |

Purity : 98.8 %

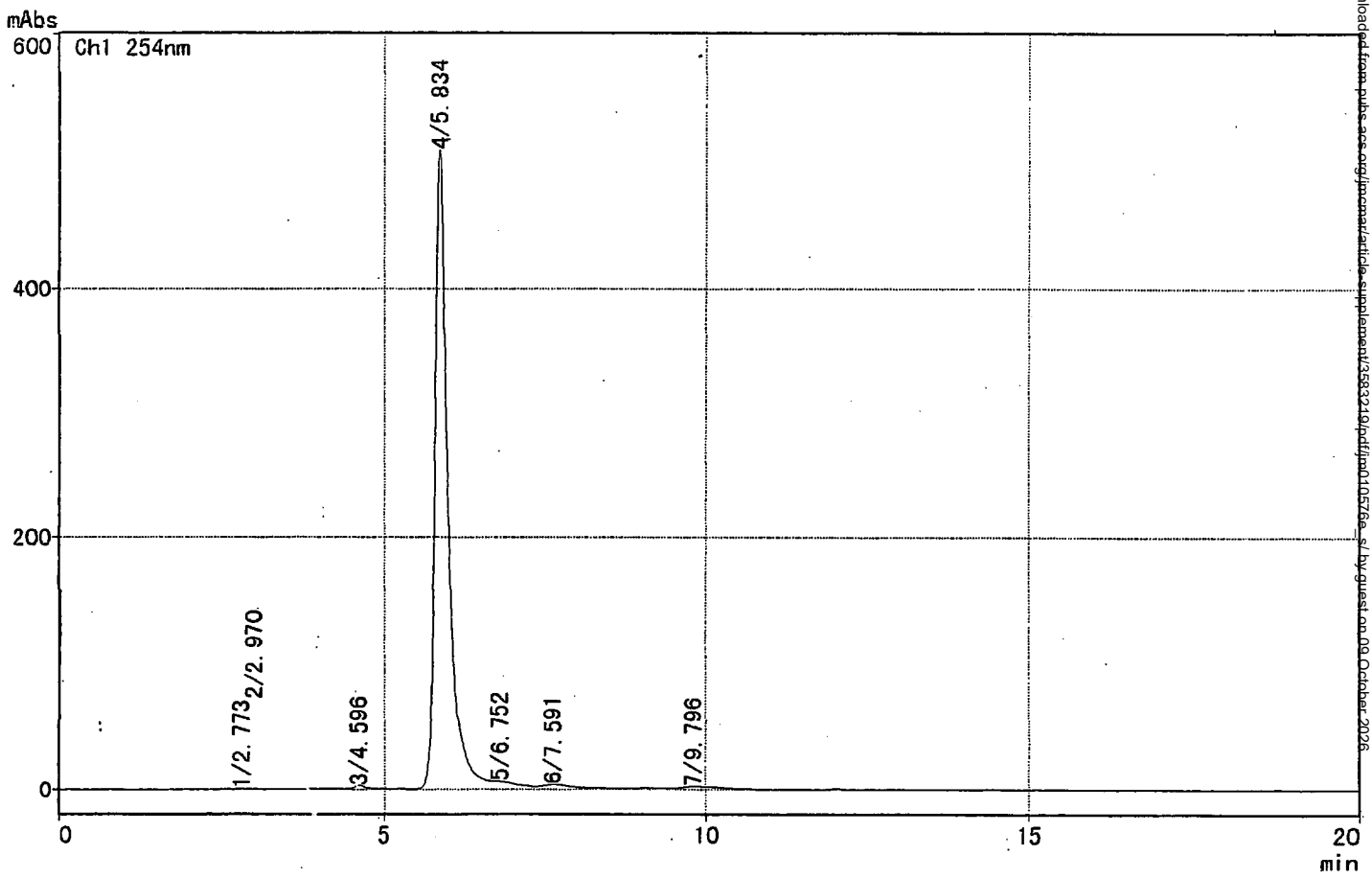
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## Condition

Column : CAPCELL PAK NH2 UG80
( SHISEIDO FINE CHEMICAL ) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 60%Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : Mitragynine pseudoindoxyl



Mitragynine pseudoindoxyl (**2**)




*** ピークレポート ***

| PKNO | ChNO | AREA | TIME |
|---|---|---|---|
| 1 | 1 | 1308 | 2.773 |
| 2 | 1 | 2538 | 2.970 |
| 3 | 1 | 28437 | 4.596 |
| 4 | 1 | 7168618 | 5.834 |
| 5 | 1 | 12388 | 6.752 |
| 6 | 1 | 50334 | 7.591 |
| 7 | 1 | 56022 | 9.796 |
| | | 7319645 | |

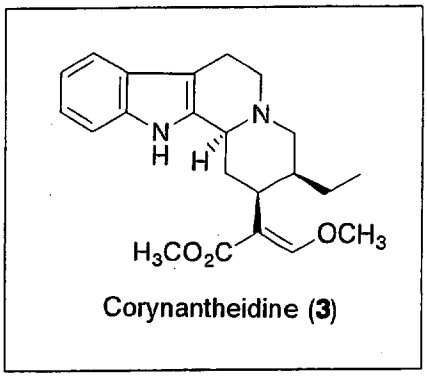
Purity : 97.9 %

Downloaded from pubs.acs.org/jmcmar/article-supplement/3583219/pdf/jm010576e_s/by guest on 09 October 2026

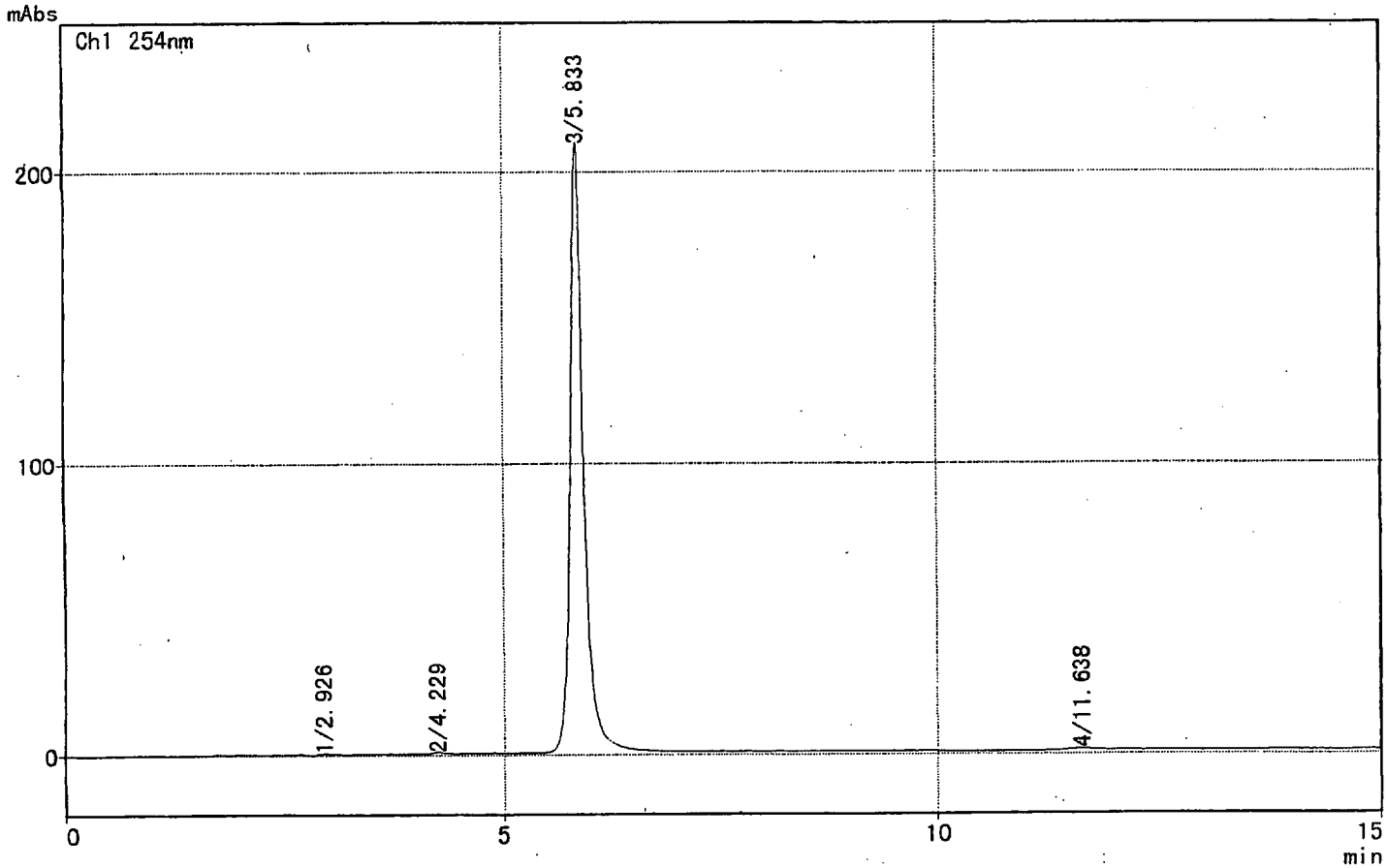
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## Condition

Column : CAPCELL PAK NH2 UG80
( SHISEIDO FINE CHEMICAL ) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 50% Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : Corynantheidine




Corynantheidine (**3**)




*** ピークレポート ***

| PKNO | ChNO | AREA | TIME |
|---|---|---|---|
| 1 | 1 | 2801 | 2.926 |
| 2 | 1 | 2870 | 4.229 |
| 3 | 1 | 2288486 | 5.833 |
| 4 | 1 | 12663 | 11.638 |
| | | 2306820 | |

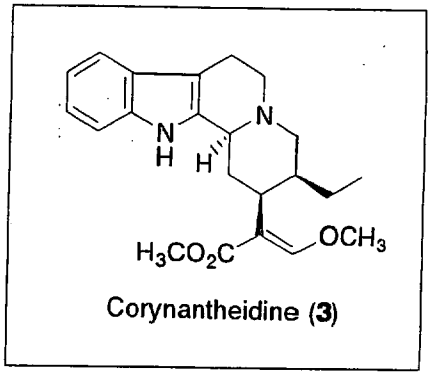
Purity : 99.2 %

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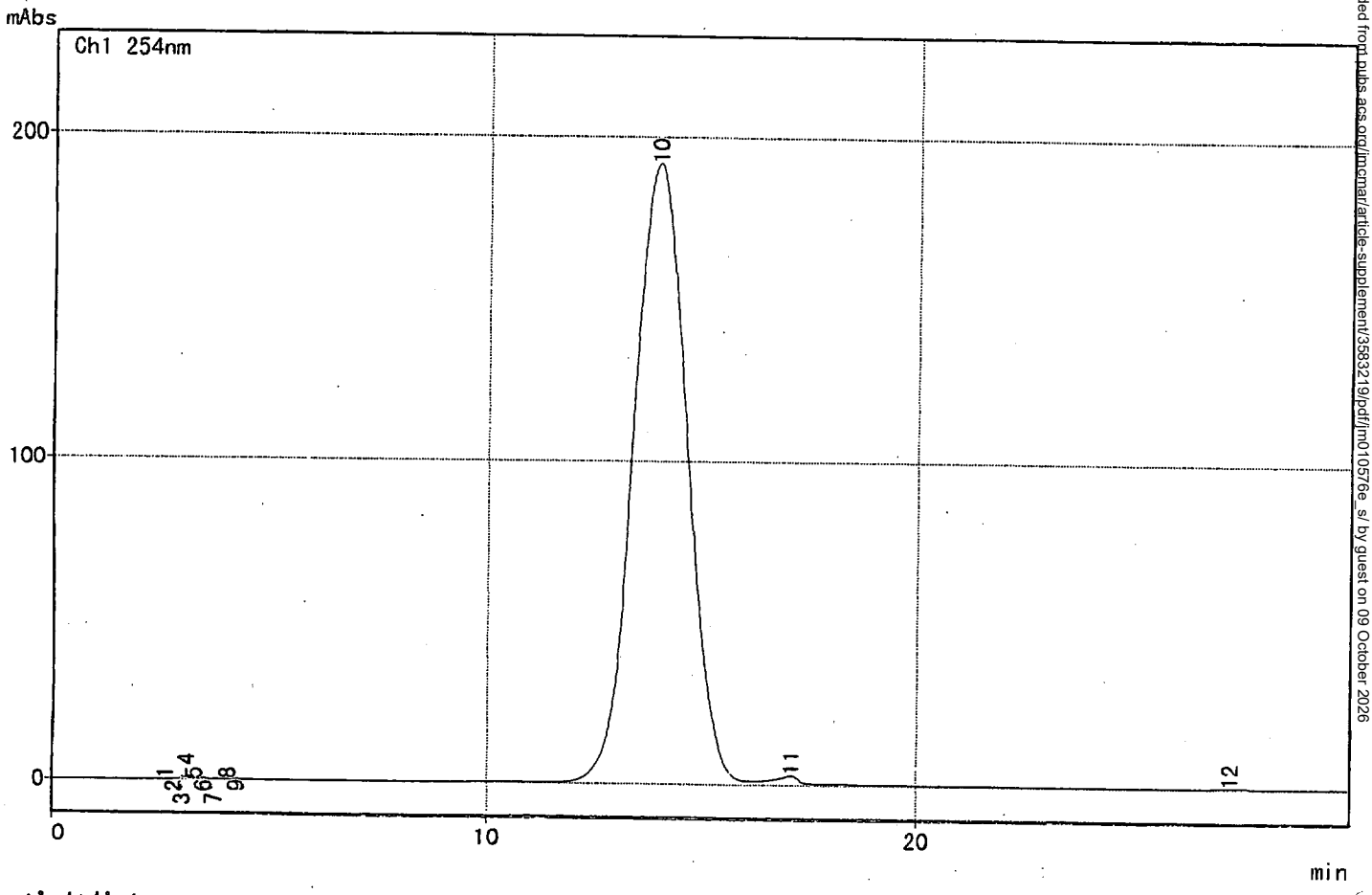
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## Condition

Column : PEGASIL Silica 60-5
( Senshu Pak ) S-5μm
Column size : 4.6mm x 250mm
Flow Rate : 1.0mL/min
Eluent : 5% MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : Corynantheidine




Corynantheidine (**3**)




*** ピークレポート ***

| PKNO | ChNO | AREA | TIME |
|---|---|---|---|
| 1 | 1 | 318 | 2.549 |
| 2 | 1 | 487 | 2.720 |
| 3 | 1 | 6073 | 2.930 |
| 4 | 1 | 2560 | 3.028 |
| 5 | 1 | 3154 | 3.219 |
| 6 | 1 | 4063 | 3.431 |
| 7 | 1 | 2599 | 3.654 |
| 8 | 1 | 664 | 3.957 |
| 9 | 1 | 1664 | 4.159 |
| 10 | 1 | 16709486 | 13.943 |
| 11 | 1 | 73537 | 17.086 |
| 12 | 1 | 9188 | 27.265 |
| | | 16813793 | |

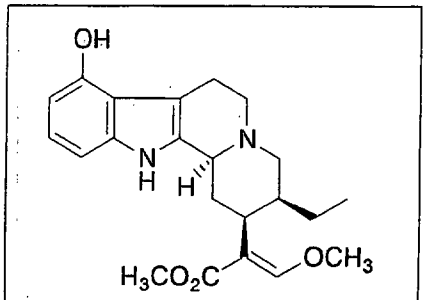
Purity : 99.4 %

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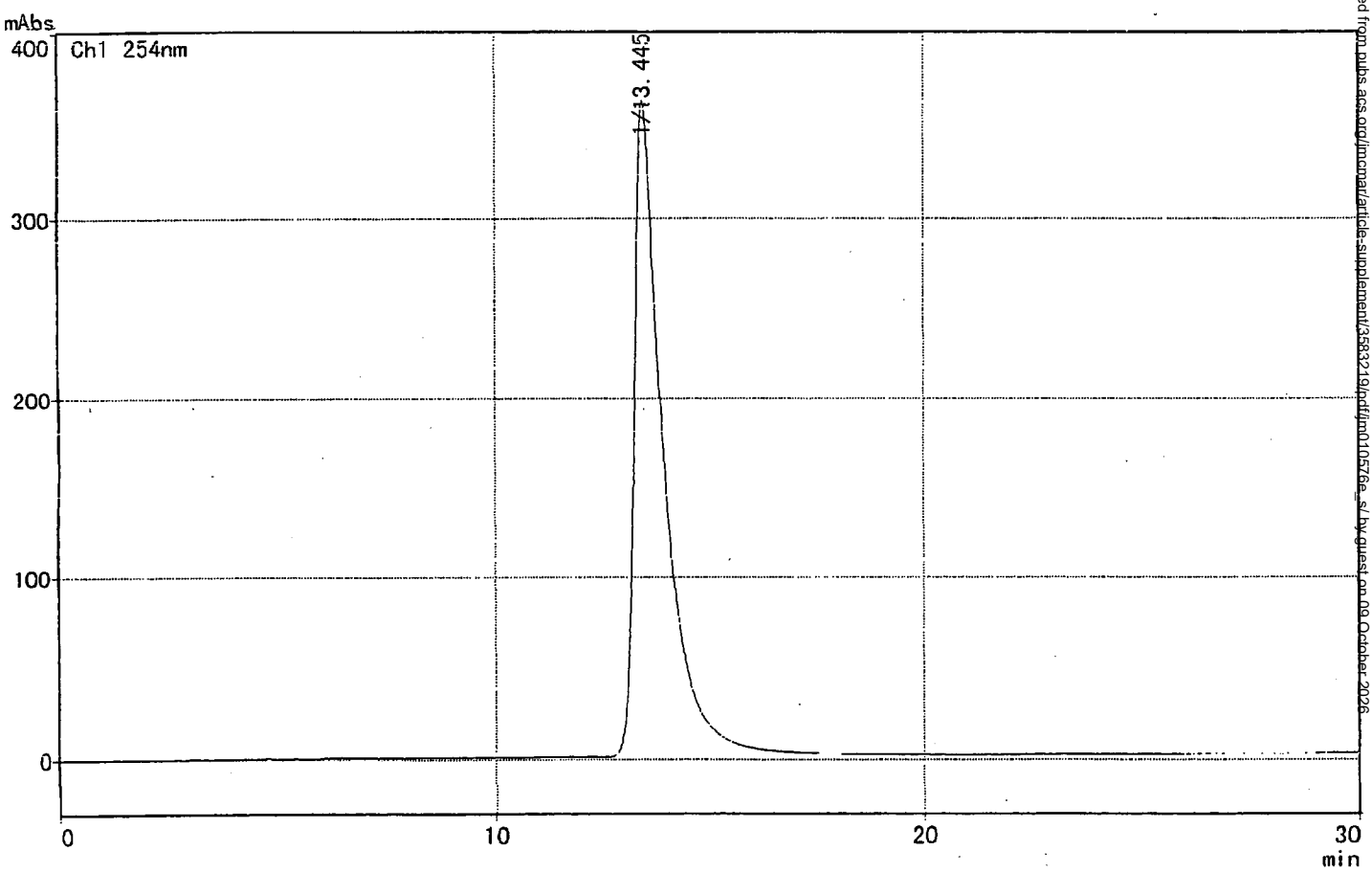
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## Condition

Column : CAPCELL PAK NH2 UG80
( SHISEIDO FINE CHEMICAL ) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 80%Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : 9-Hydroxycorynantheidine



9--Hydroxycorynantheidine (**5**)




*** ピークレポート ***

| PKNO | ChNO | AREA | TIME |
|---|---|---|---|
| 1 | 1 | 16766512 | 13.445 |
| | | 16766512 | |

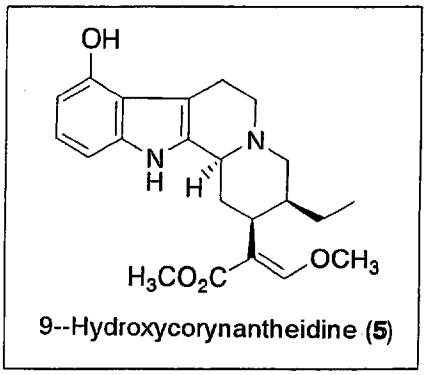
Purity : 100 %

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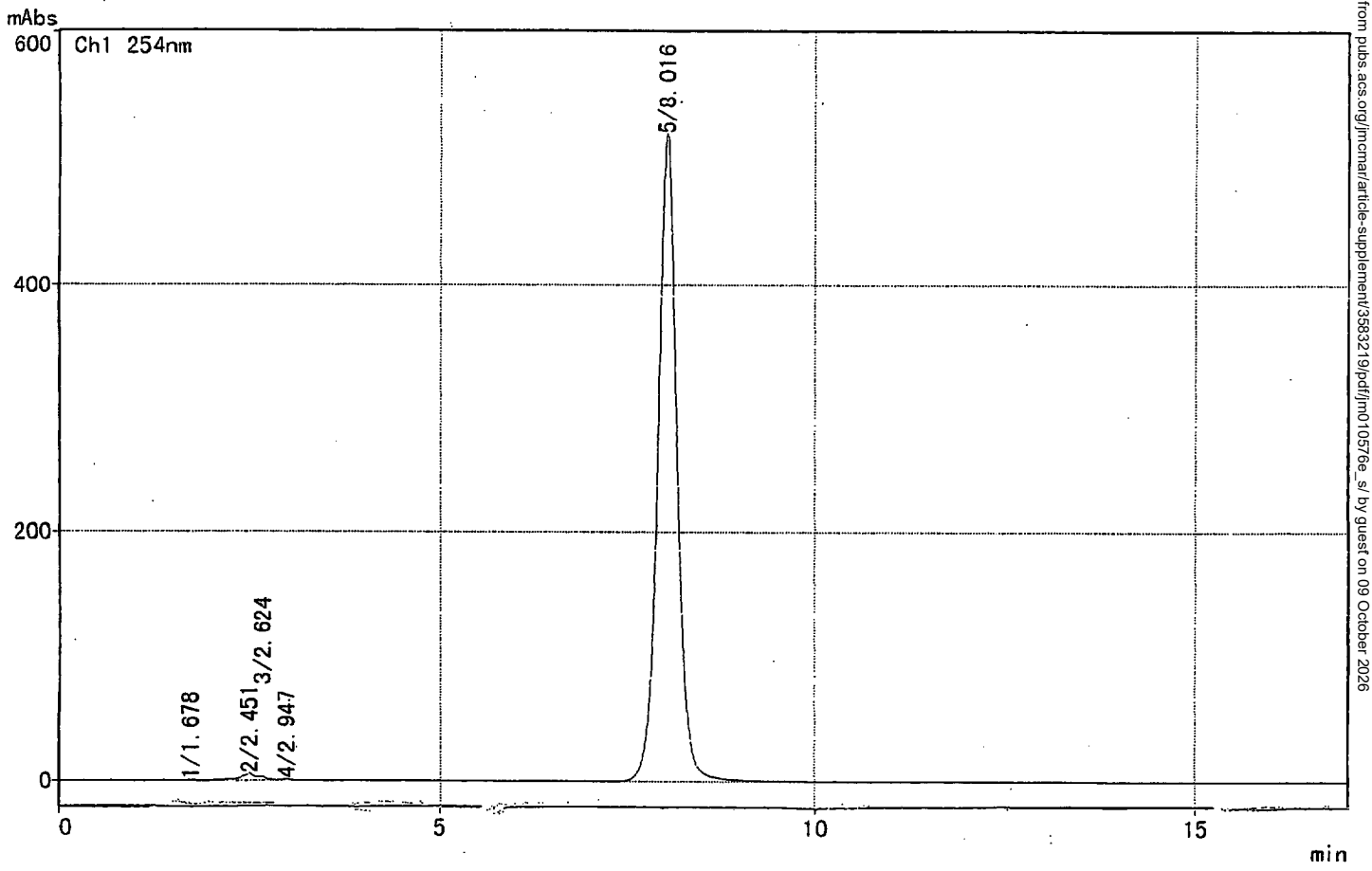
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## Condition

Column : CAPCELL PAK NH2 UG80
( SHISEIDO FINE CHEMICAL ) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 1%MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : 9-Hydroxycorynantheidine



9--Hydroxycorynantheidine (**5**)




*** ピークレポート ***

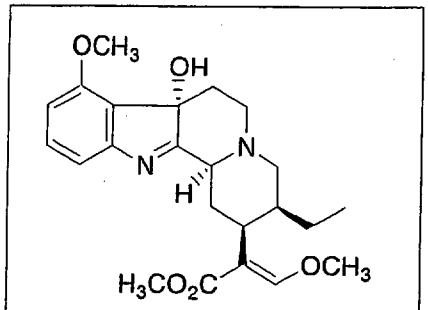
| PKNO | ChNO | AREA | TIME |
|---|---|---|---|
| 1 | 1 | 1015 | 1.678 |
| 2 | 1 | 62202 | 2.451 |
| 3 | 1 | 22885 | 2.624 |
| 4 | 1 | 8151 | 2.947 |
| 5 | 1 | 9187358 | 8.016 |
| | | 9281611 | |

Purity : 99.0 %

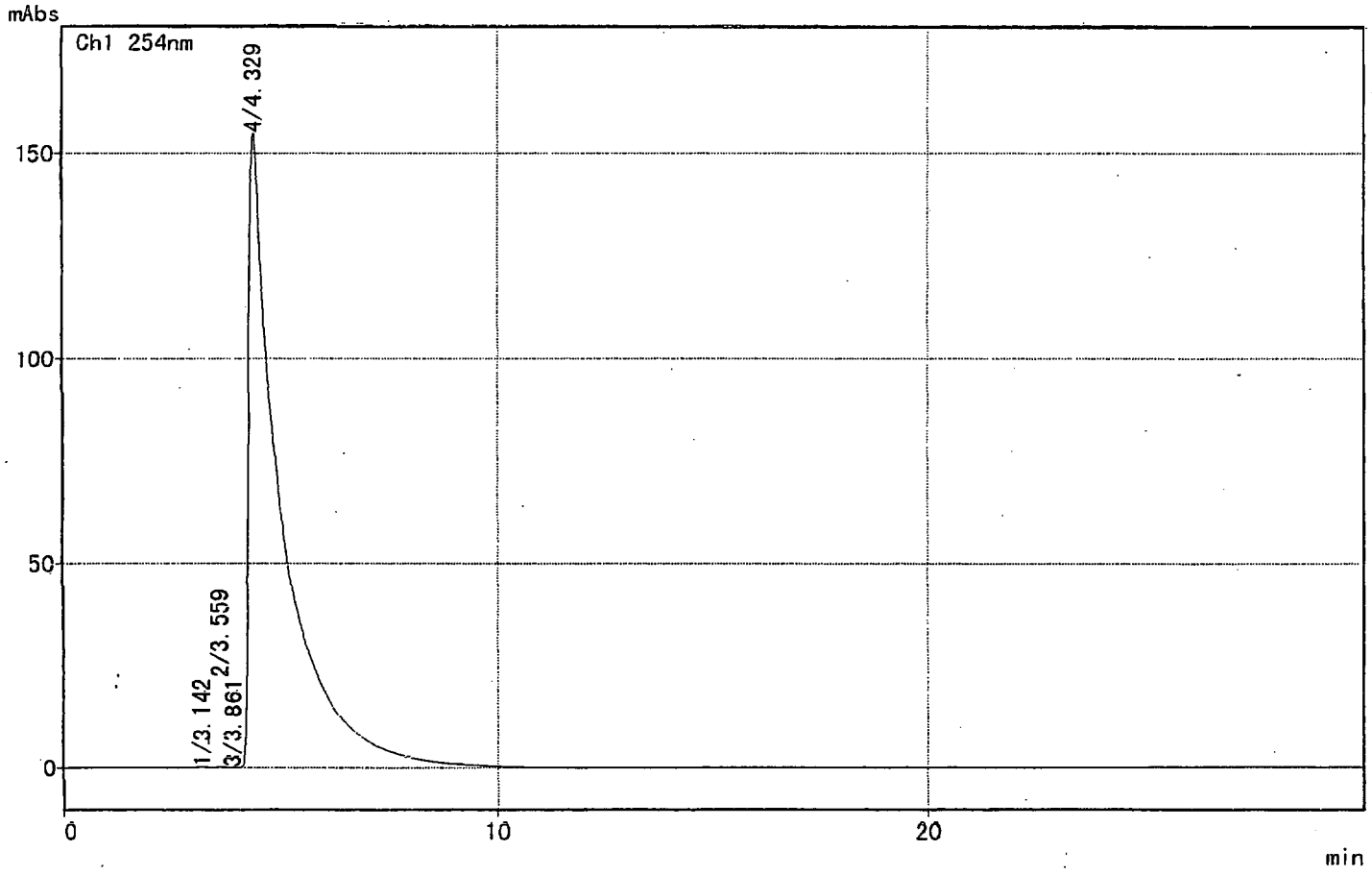
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## Condition

Column : YMC-Pack A-012 SIL
( YMC ) S-5μm
Column size : 6.0mm x 150mm
Flow Rate : 1.0mL/min
Eluent : 5% MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : 7-Hydroxymitragynine




7-Hydroxymitragynine (**12**)




*** ピークレポート ***

| PKNO | ChNO | AREA | TIME |
|---|---|---|---|
| 1 | 1 | 3085 | 3.142 |
| 2 | 1 | 2716 | 3.559 |
| 3 | 1 | 108 | 3.861 |
| 4 | 1 | 8119304 | 4.329 |
| | | 8125213 | |

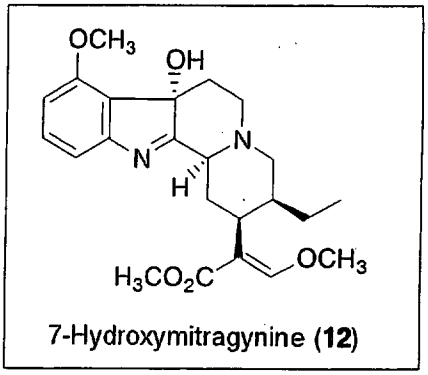
Purity : 99.9 %

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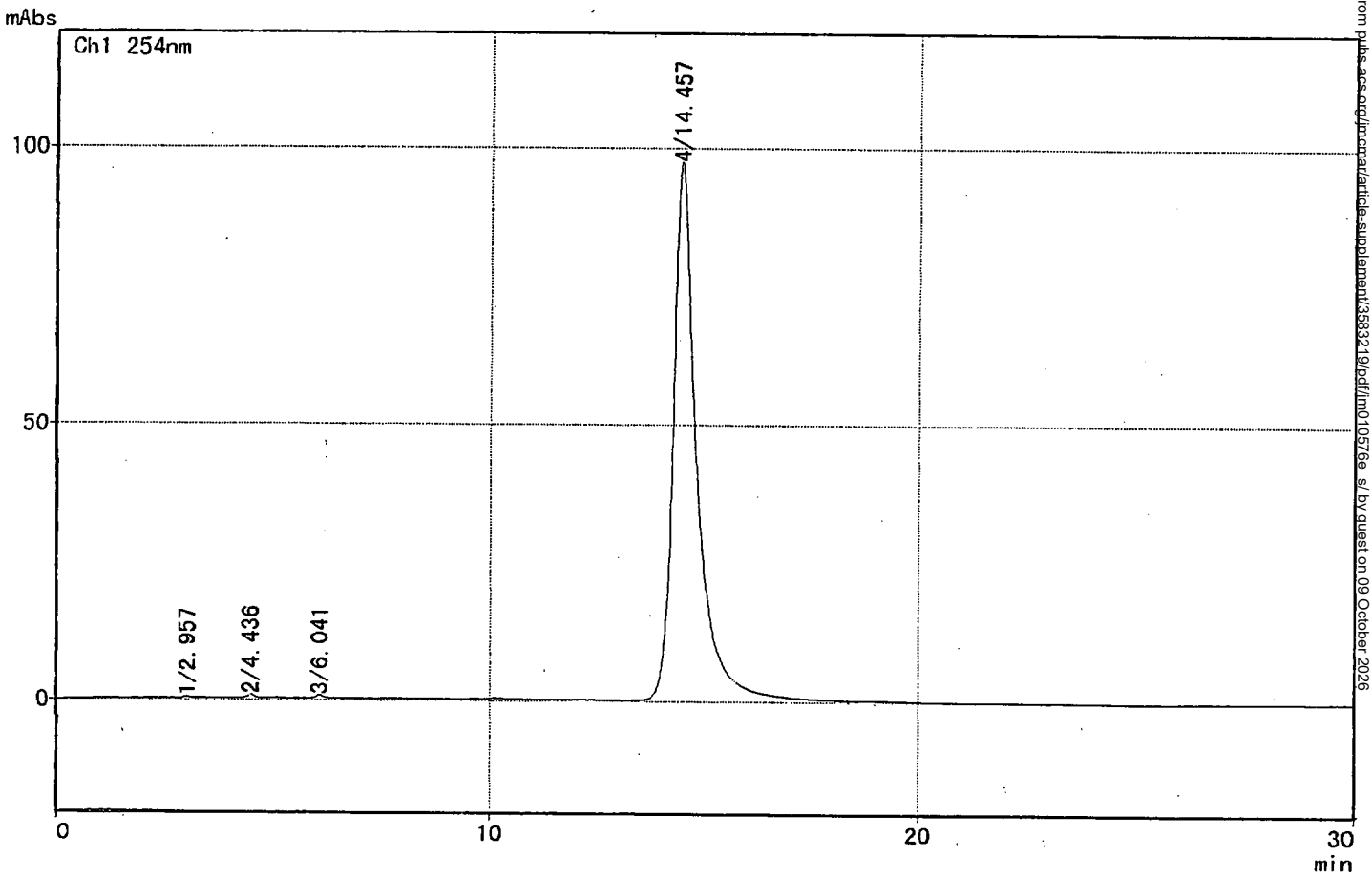
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## Condition

Column : CAPCELL PAK NH2 UG80
( SHISEIDO FINE CHEMICAL ) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 50% Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : 7-Hydroxymitragynine



7-Hydroxymitragynine (**12**)



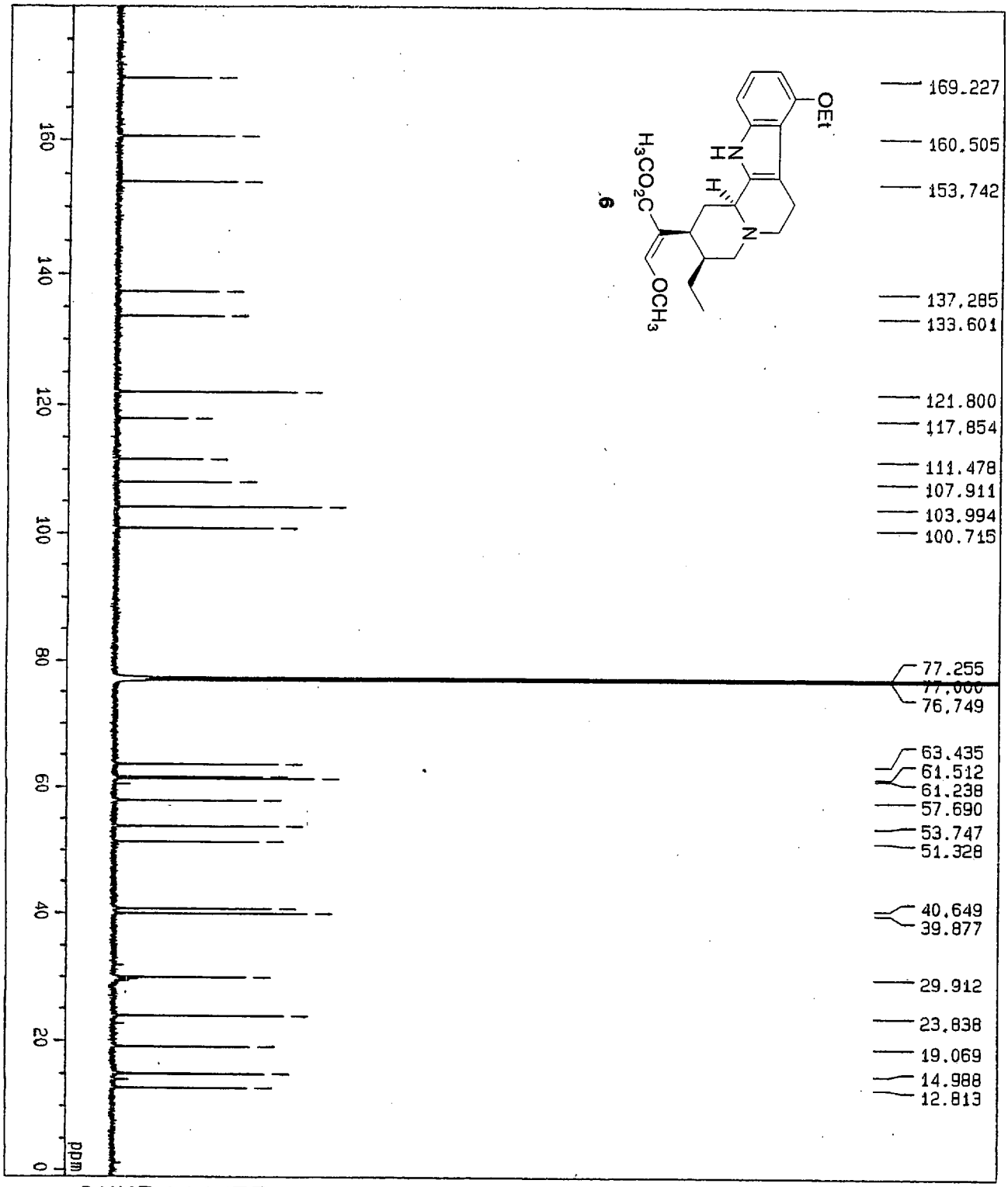

*** ピークレポート ***

| PKNO | ChNO | AREA | TIME |
|---|---|---|---|
| 1 | 1 | 1980 | 2.957 |
| 2 | 1 | 5371 | 4.436 |
| 3 | 1 | 5748 | 6.041 |
| 4 | 1 | 3606465 | 14.457 |
| | | 3619563 | |

Purity : 99.6 %

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7-MAY-1999 18:21:40.32
DFILE : ALPHA
SFILE : SKISHIKAWA-MG2-1-1BCM
COMNT : MG-2-1-1 BCM
EXMOD : SINGL
IRMOD : BCM
POINT : 65536
FREQU : 30030.03 Hz
SCANS : 1200
DUMMY : 4
ACQTM : 2.1823 sec
PD : 1.5000 sec
RGAIN : 23

PW1 : 4.40 usec

OBNUC : 13C
OBFRQ : 125.65 MHz
OBSET : 127782.65 Hz

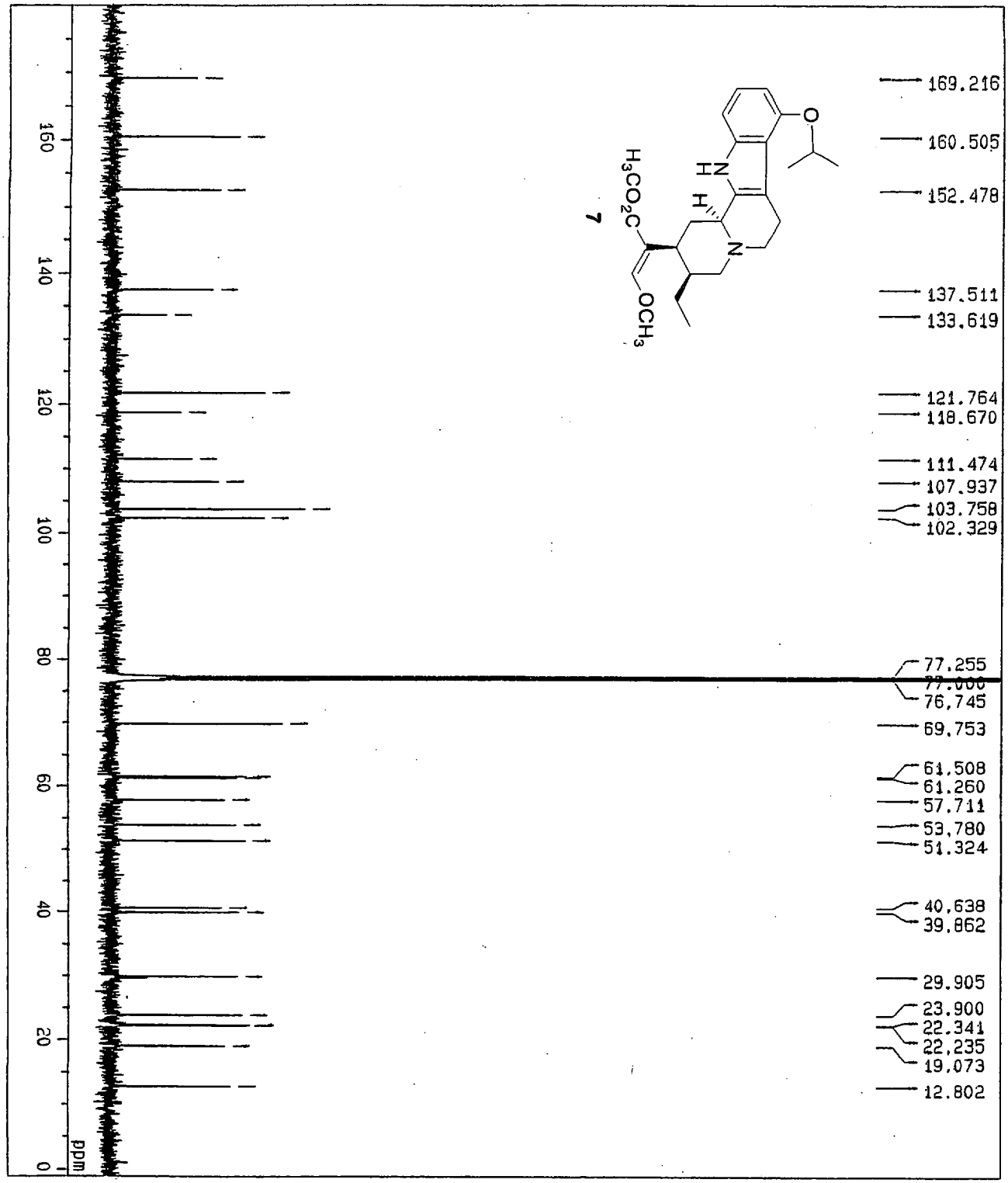
IRNUC : 1H
IRFRQ : 500.00 MHz
IRSET : 162410.00 Hz
IRATN : 511
IRAPW : 67.0 usec
IRBP1 : 30
IRBP2 : 6
IRNS : 0

ADBIT : 16
CTEMP : 20.5 c
CSPED : 11 Hz
SLVNT : CDCL3

RESOL : 0.46 Hz
BF : 2.00 Hz
T1 : 0.00 %
T2 : 0.00 %
T3 : 90.00 %
T4 : 100.00 %
REFVL : 77.00 ppm
XE : 22763.09 Hz
XS : 1212.68 Hz
operator :

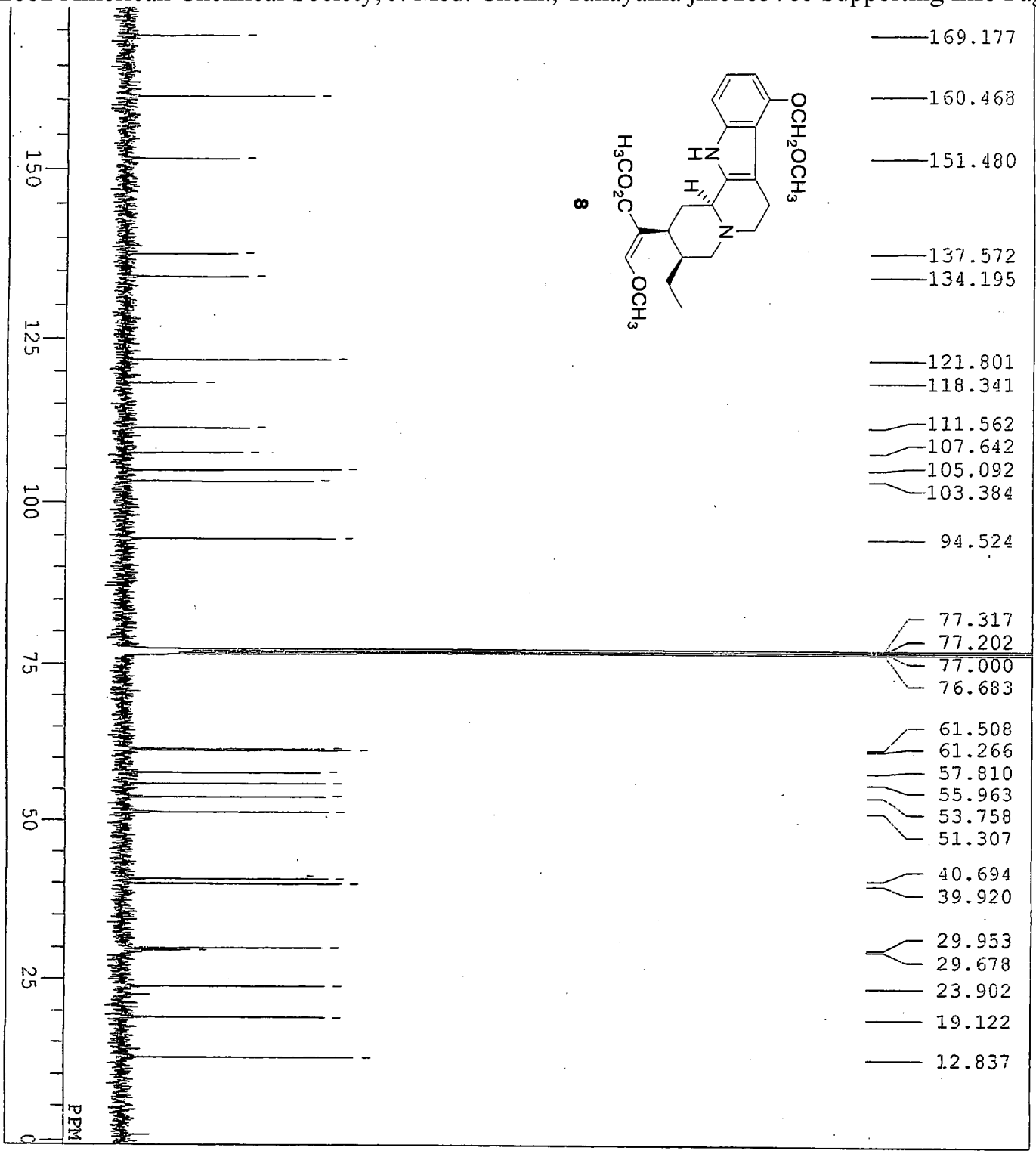
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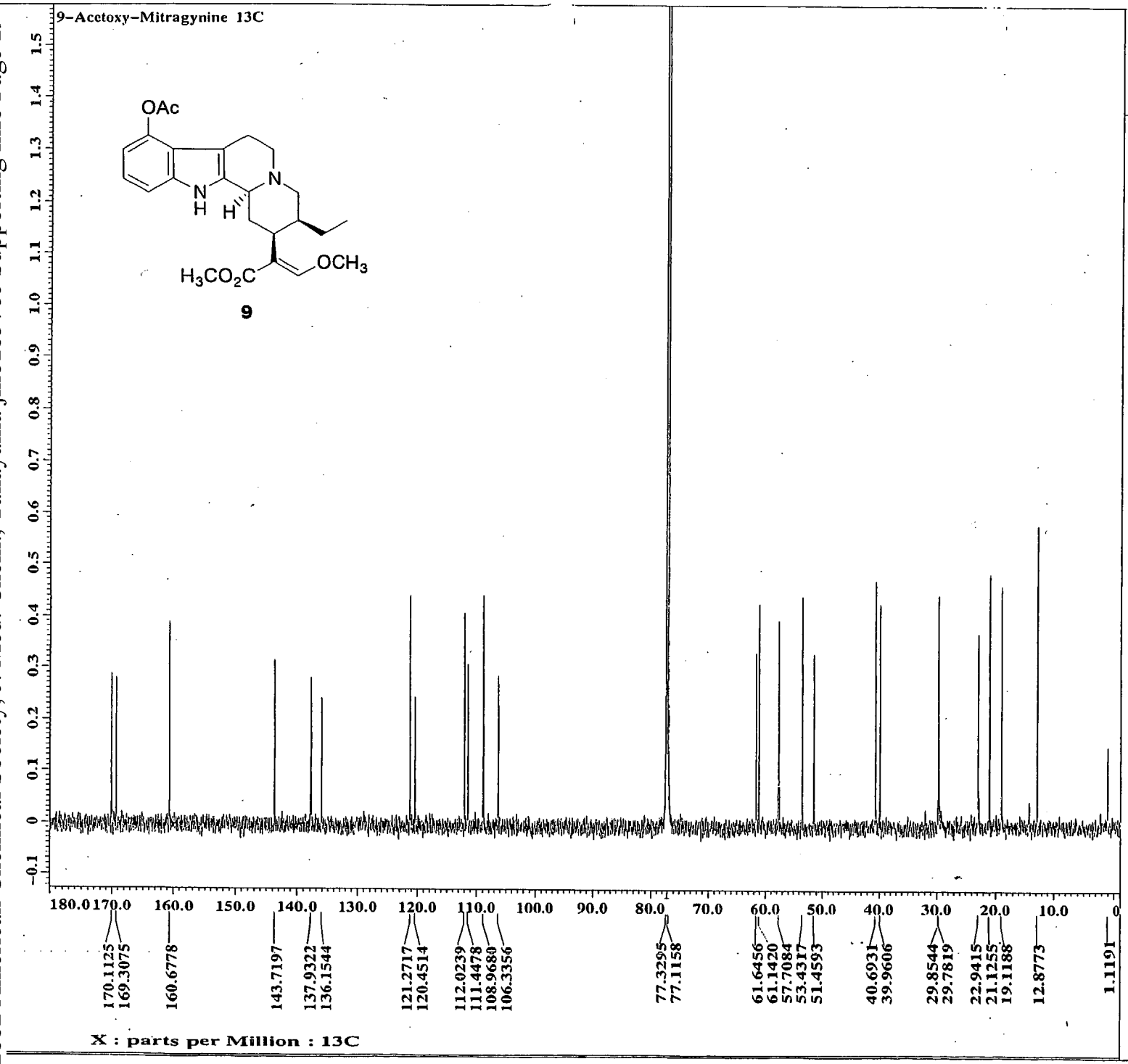



```
DFILE   D:\SK\ISHIKAWA\SKHI-MG1
COMNT   MG10-2-1BCM
DATIM
OBNUC   13C
EXMOD   SINGL-BCM
OBFRQ     100.40 MHz
OBSET     130.00 KHz
OBFIN    5500.0 Hz
POINT      65536
FREQU   27100.3 Hz
SCANS        1120
ACQTM      2.418 sec
PD         1.300 sec
PW1          4.8 us
IRNUC   1H
CTEMP     29.6 c
SLVNT   CDCL3
EXREF     77.00 ppm
BF         2.00 Hz
RGAIN        23
```

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JEOL

---- ACQUISITION PARAMETERS ----

| | |
|---|---|
| File Name | = SKISHIKAWA-9-OAC-MIT |
| Author | = JEOL LTD. |
| Sample ID | = SKISHIKAWA-9-OAC-MIT |
| Content | = 9-Acetoxy-Mitragynin |
| Creation Date | = 1-MAY-2001 19:03:07 |
| Revision Date | = 1-MAY-2001 19:14:59 |
| Spec Site | = ECP600 |
| Spec Type | = DELTA_NMR |
| Data Format | = 1D COMPLEX |
| Dimensions | = X |
| Dim Title | = 13C |
| Dim Size | = 65536 |
| Dim Units | = [ppm] |
| Actual_start_time | = 1-MAY-2001 19:03:08 |
| Delay_of_start | = 1[s] |
| Digital_filter | = FALSE |
| End_time | = 1-MAY-2001 19:37:41 |
| Experiment | = single_pulse_bil |
| Field_strength | = 14.09636928[T] |
| Filter_mode | = BUTTERWORTH |
| Filter_width | = 18.8641788[kHz] |
| Irr_code | = 194 |
| Irr_domain | = 1H |
| Irr_freq | = 600.17530548[MHz] |
| Irr_noise | = WAUGH |
| Irr_offset | = 5.0[ppm] |
| Irr_pwidth | = 63[us] |
| Iterations | = 1 |
| Local_time | = 1-MAY-2001 19:03:06 |
| Obs_noise | = WAUGH |
| Obs_pwidth | = 1[us] |
| Probe_id | = 2692 |
| Recvr_gain | = 27 |
| Relaxation_delay | = 1.5[s] |
| Scans | = 960 |
| Solvent | = CHLOROFORM-D |
| Spin_get | = 14[Hz] |
| Spin_lock_90 | = 45[us] |
| Spin_lock_attn | = 11.5[dB] |
| Temp_get | = 25.2[dC] |
| X90 | = 10.8[us] |
| X_acq_duration | = 1.736704[s] |
| X_angle | = 30[deg] |
| X_domain | = 13C |
| X_freq | = 150.9300327[MHz] |
| X_offset | = 110[ppm] |
| X_points | = 65536 |
| X_prescans | = 4 |
| X_pulse | = 3.6[us] |
| X_resolution | = 0.57581215[Hz] |
| X_sweep | = 37.73584906[kHz] |
| Tri90 | = 10[us] |
| Tri_noise | = WALTZ |
| Tri_pwidth | = 1[us] |
| Qua90 | = 10[us] |
| Qua_noise | = WAUGH |
| Qua_pwidth | = 1[us] |

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MITR N-O BCM2




Mitragynine *N*-oxide (**10**)

JEOL

2-FEB-1996 09:03:43.71

```
DFILE : ALPHA
SFILE : SKKMN021BCM

COMNT : MITR N-O BCM2
EXMOD : SINGL
IRMOD : BCM
POINT :     32768
FREQU : 27100.27 Hz
SCANS :      3480
DUMMY :         4
ACQTM :    1.2091 sec
PD    :    0.7909 sec
RGAIN :        23

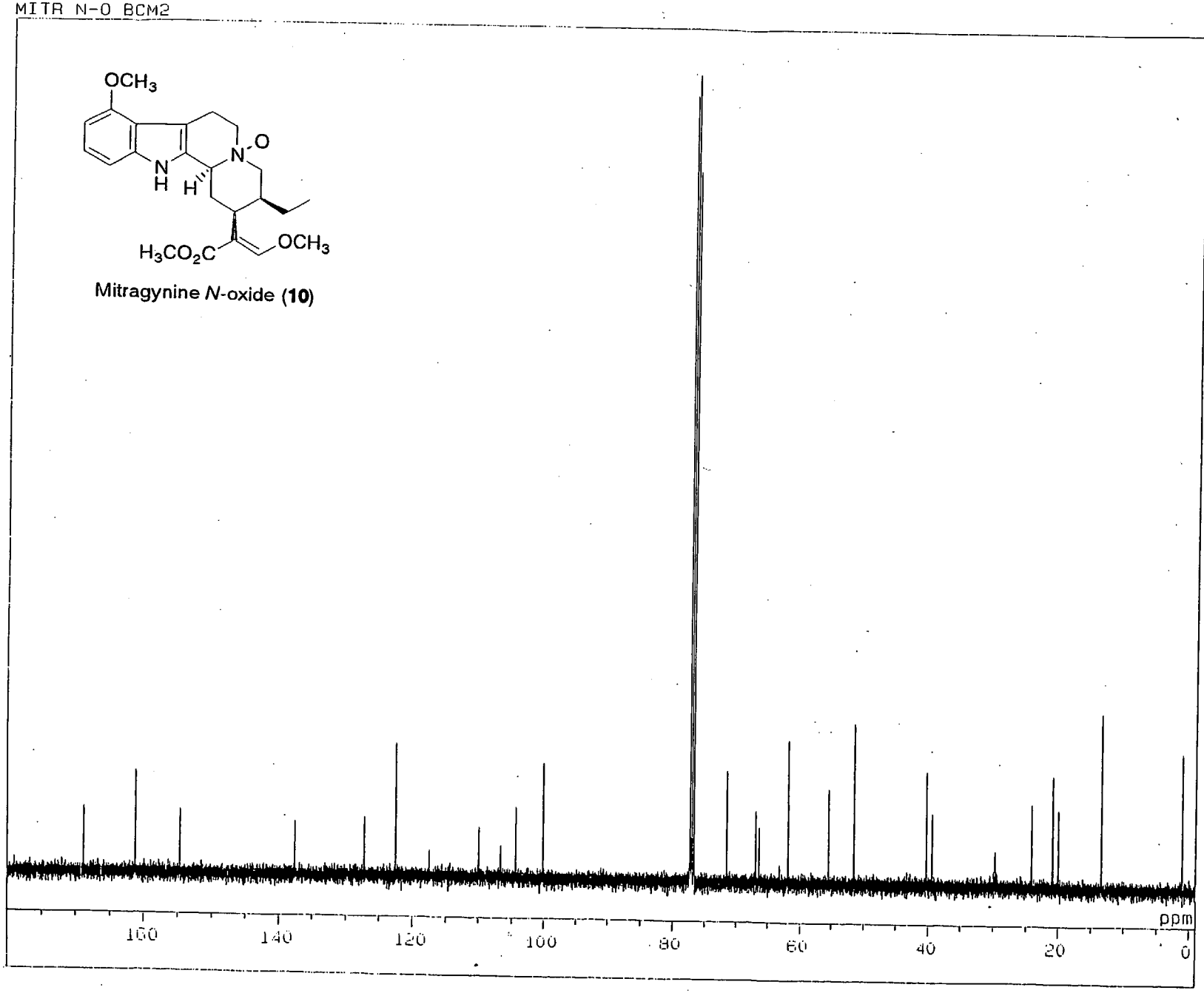
PW1   :      4.55 usec


OBNUC : 13C
OBFRQ :    100.40 MHz
OBSET : 135500.00 Hz

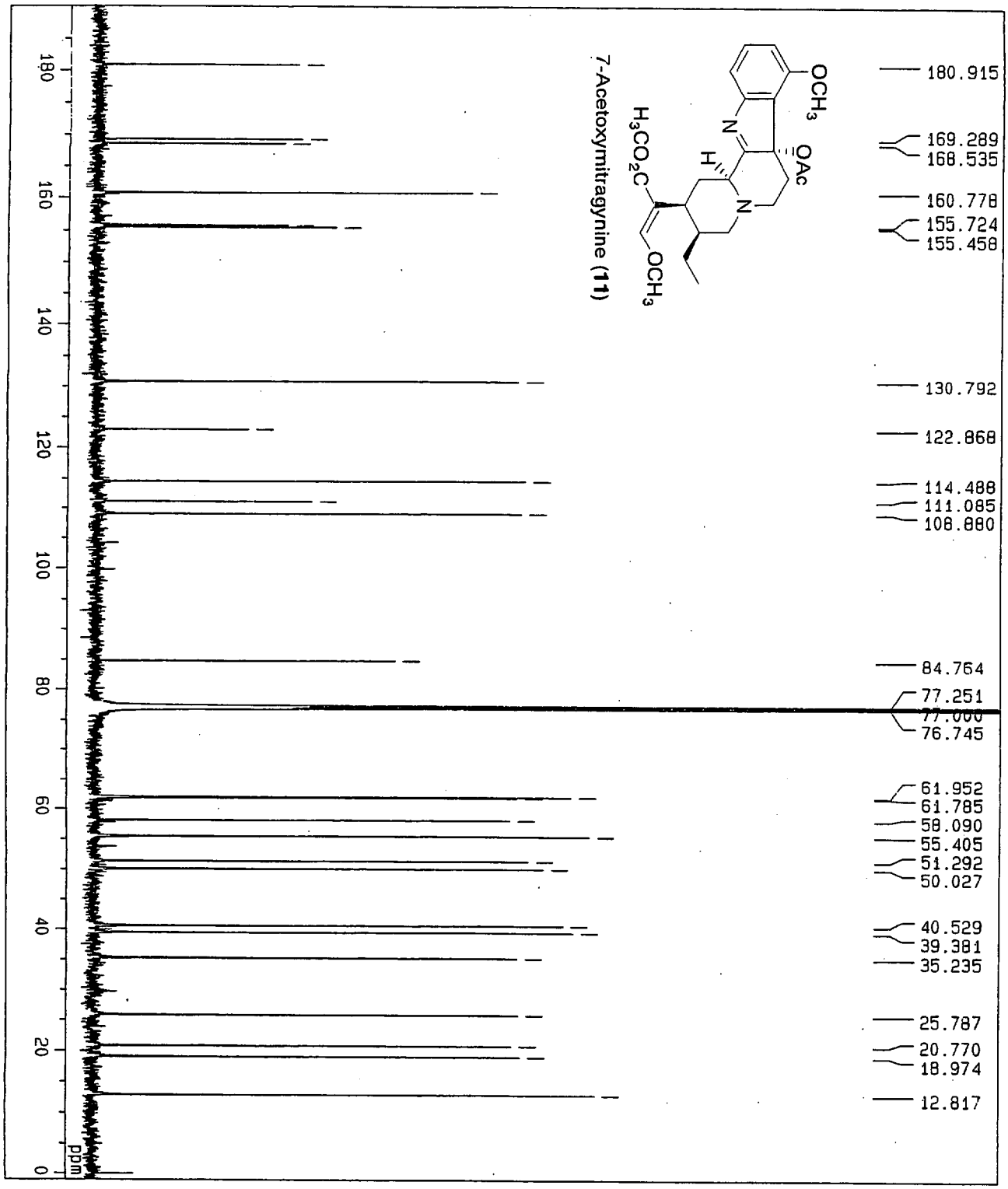
IRNUC : 1H
IRFRQ :    399.65 MHz
IRSET : 134500.00 Hz
IRATN :       511
IRRPW :      45.0 usec
IRBP1 :        25
IRBP2 :         6
IRANS :         0

ADBIT :        16
CTEMP :      25.0 c
CSPED :         9 Hz
SLVNT : CDCL3

RESOL :      0.83 Hz
BF    :      3.00 Hz
T1    :      0.00 %
T2    :      0.00 %
T3    :     90.00 %
T4    :    100.00 %
REFVL :     77.00 ppm
XE    : 18193.93 Hz
XS    :  1111.95 Hz
operator
```



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```
5-JUN-2001 20:14:32.01
DFILE : ALPHA
SFILE : SKHI-7-OACMIT-BCM
COMNT : 7-OACMIT-BCM
EXMOD : SINGL
IRMOD : BCM
POINT :          65536
FREQU :   30030.03 Hz
SCANS :           2800
DUMMY :              4
ACQTM :     2.1823 sec
PD    :     1.5000 sec
RGAIN :             23
PW1   :     3.95 usec

OBNUC : 13C
OBFRQ :    125.65 MHz
OBSET : 127782.65 Hz

IRNUC : 1H
IRFRQ :    500.00 MHz
IRSET : 162410.00 Hz
IRATN :           511
IRRPW :      50.0 usec
IRBP1 :            30
IRBP2 :             5
IRRNS :             0

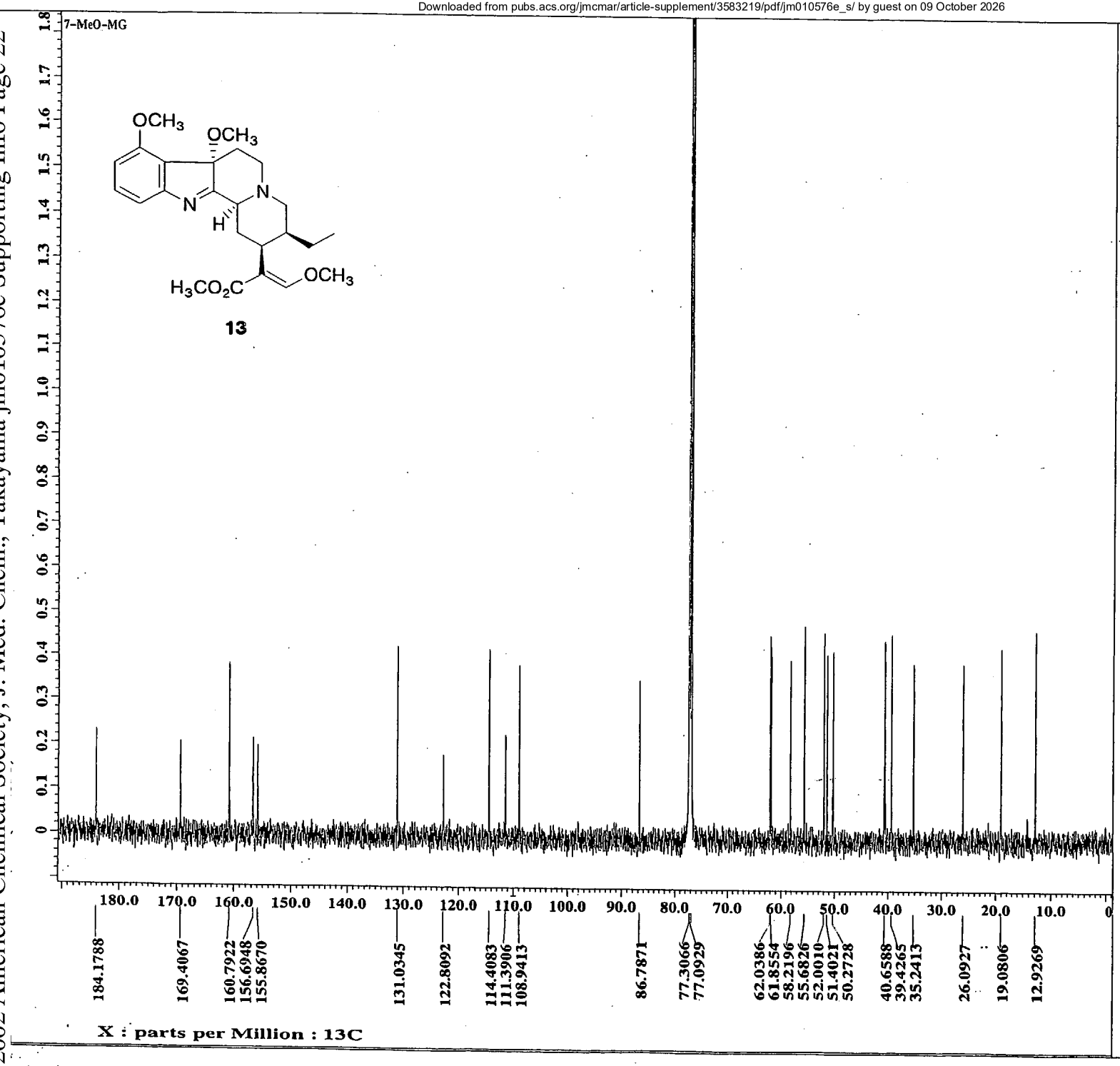
ADBIT :            16
CTEMP :       21.9 c
CSPED :       13 Hz
SLVNT : CDCL3

RESOL :      0.46 Hz
BF    :      3.00 Hz
T1    :      0.00 %
T2    :      0.00 %
T3    :     90.00 %
T4    :    100.00 %
REFVL :     77.00 ppm
XE    : 24020.91 Hz
XS    :    584.69 Hz
operator
```

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JEOL

```
---- ACQUISITION PARAMETERS ----
File Name            = SKISHIKAWA-7MEO-MG-1
Author               = JEOL LTD.
Sample ID            = SKISHIKAWA-7MEO-MG
Content              = 7-MeO-MG
Creation Date        = 30-OCT-2001 14:27:14

Revision Date        = 30-OCT-2001 14:34:16
Spec Site            = ECP600

Spec Type            = DELTA_NMR
Data Format          = 1D COMPLEX
Dimensions           = X
Dim Title            = 13C
Dim Size             = 65536
Dim Units            = [ppm]
Actual_start_time    = 30-OCT-2001 14:27:13
Delay_of_start       = 1[s]
Digital_filter       = FALSE
End_time             = 30-OCT-2001 15:18:57
Experiment           = single_pulse_bil
Field_strength       = 14.09636928[T]
Filter_mode          = BUTTERWORTH
Filter_width         = 18.8641788[kHz]
Irr_code             = 194
Irr_domain           = 1H
Irr_freq             = 600.17530548[MHz]
Irr_noise            = WAUGH
Irr_offset           = 5.0[ppm]
Irr_pwidth           = 60[us]
Iterations           = 1
Local_time           = 30-OCT-2001 14:27:13
Obs_noise            = WAUGH
Obs_pwidth           = 1[us]
Probe_id             = 2692
Recvr_gain           = 26
Relaxation_delay     = 1.5[s]
Scans                = 1842
Solvent              = CHLOROFORM-D
Spin_get             = 15[Hz]
Spin_lock_90         = 45[us]
Spin_lock_attn       = 11.5[dB]
Temp_get             = 24.1[dC]
X90                  = 13[us]
X_acq_duration       = 1.736704[s]
X_angle              = 30[deg]
X_domain             = 13C
X_freq               = 150.9300327[MHz]
X_offset             = 110[ppm]
X_points             = 65536
X_prescans           = 4
X_pulse              = 4.33333333[us]
X_resolution         = 0.57581215[Hz]
X_sweep              = 37.73584906[kHz]
Tri90                = 10[us]
Tri_noise            = WALTZ
Tri_pwidth           = 1[us]
Qua90                = 10[us]
Qua_noise            = WAUGH
Qua_pwidth           = 1[us]
```

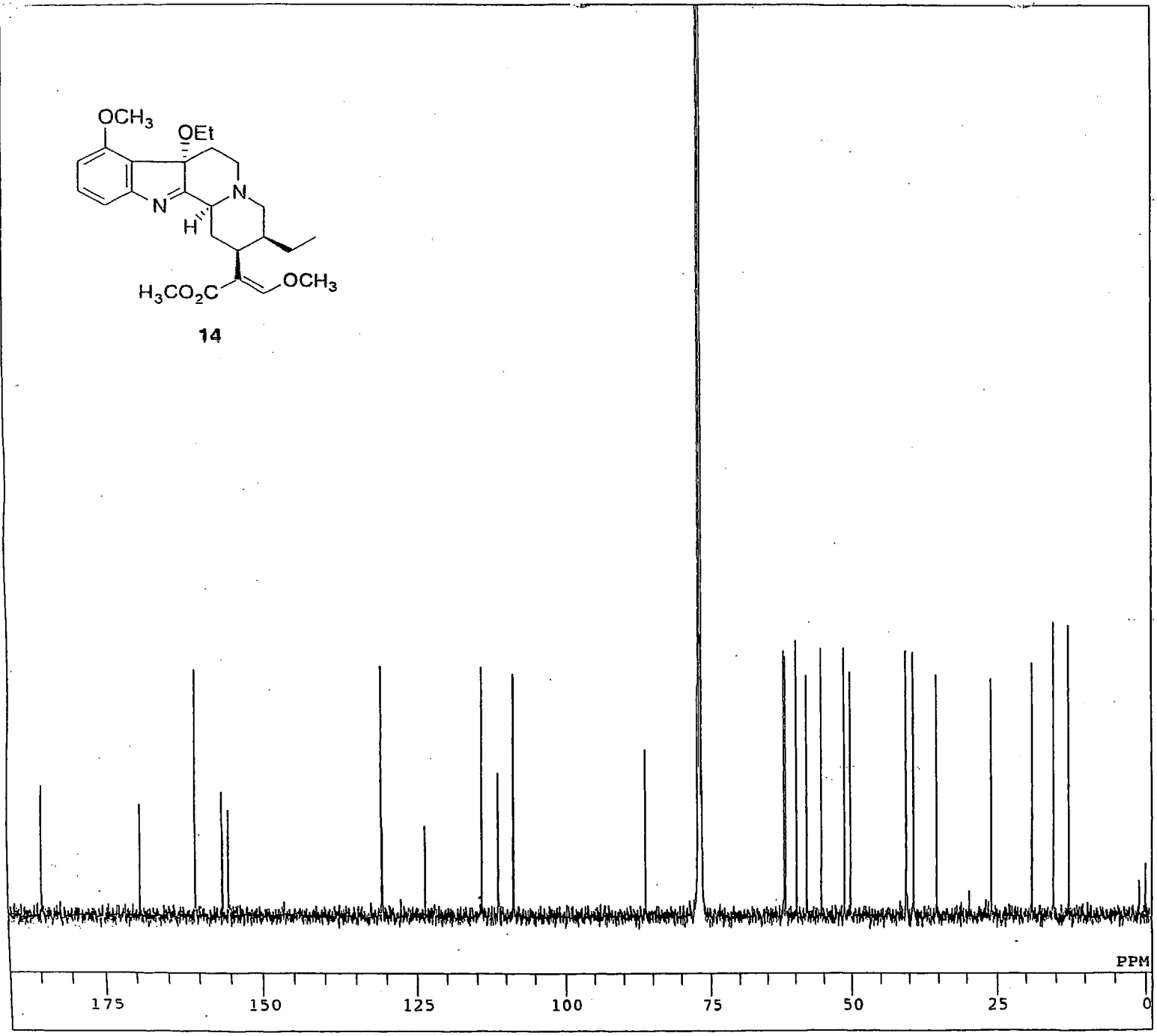
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**14**




| | |
|---|---|
| DFILE | D:\NMR-datas\Ishikawa\SKH |
| COMNT | 7-OETMIT |
| DATIM | |
| OBNUC | 13C |
| EXMOD | SINGL-BCM |
| OBFRQ | 100.40 MHz |
| OBSET | 130.00 KHz |
| OBFIN | 5500.0 Hz |
| POINT | 65536 |
| FREQU | 27100.3 Hz |
| SCANS | 12400 |
| ACQTM | 2.418 sec |
| PD | 1.300 sec |
| PW1 | 4.9 us |
| IRNUC | 1H |
| CTEMP | 23.3 c |
| SLVNT | CDCL3 |
| EXREF | 77.00 ppm |
| BF | 2.00 Hz |
| RGAIN | 23 |

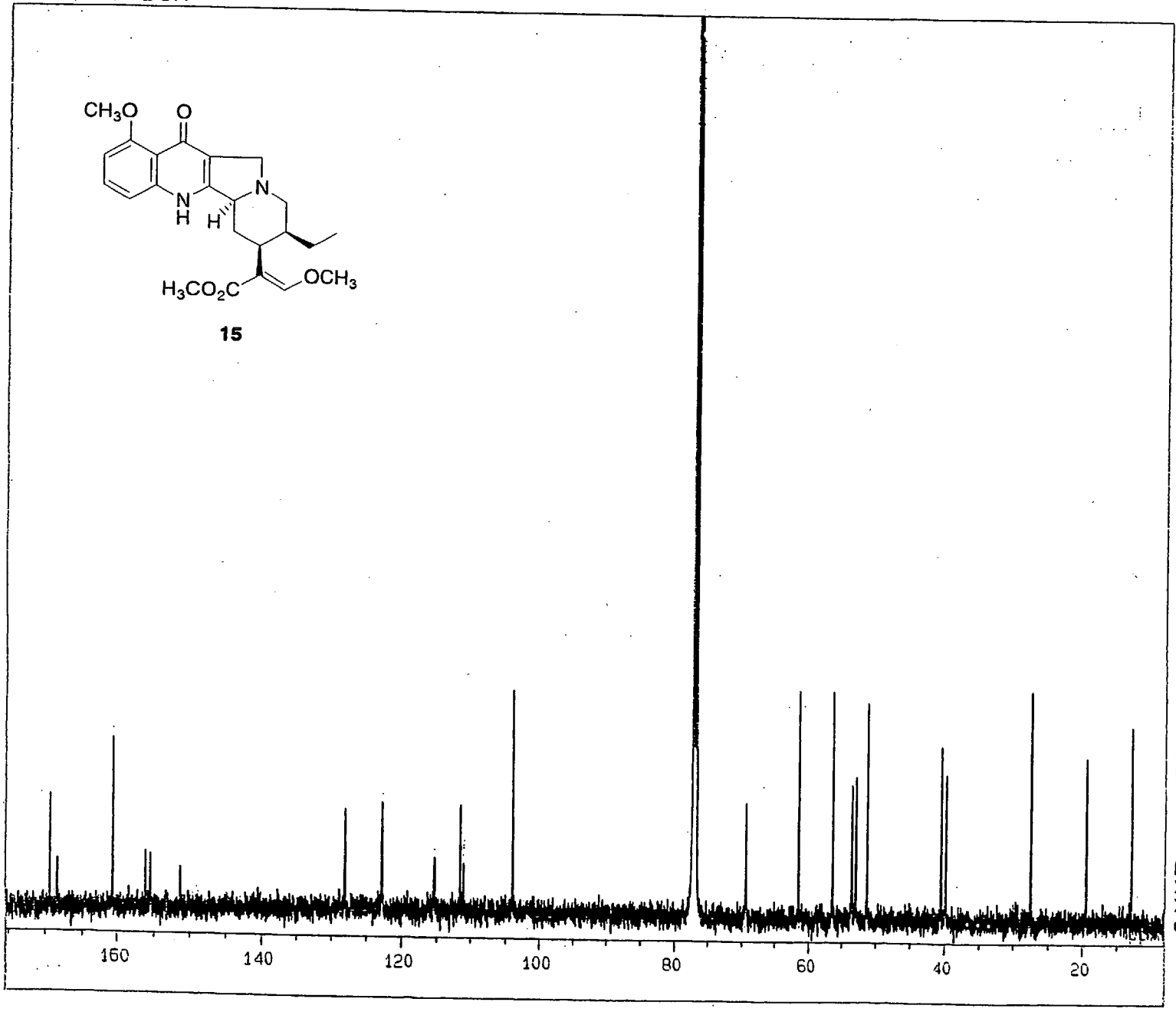
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NB1-2-1 BCM




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DFILE : ALPHA
SFILE : SKKATO-NB1-2-1BCM

COMNT : NB1-2-1 BCM
EXMOD : SINGL
IRMOD : BCM
POINT : 65536
FREQU : 30030.03 Hz
SCANS : 2710
DUMMY : 4
ACQTM : 2.1823 sec
PD : 1.5000 sec
RGAIN : 23

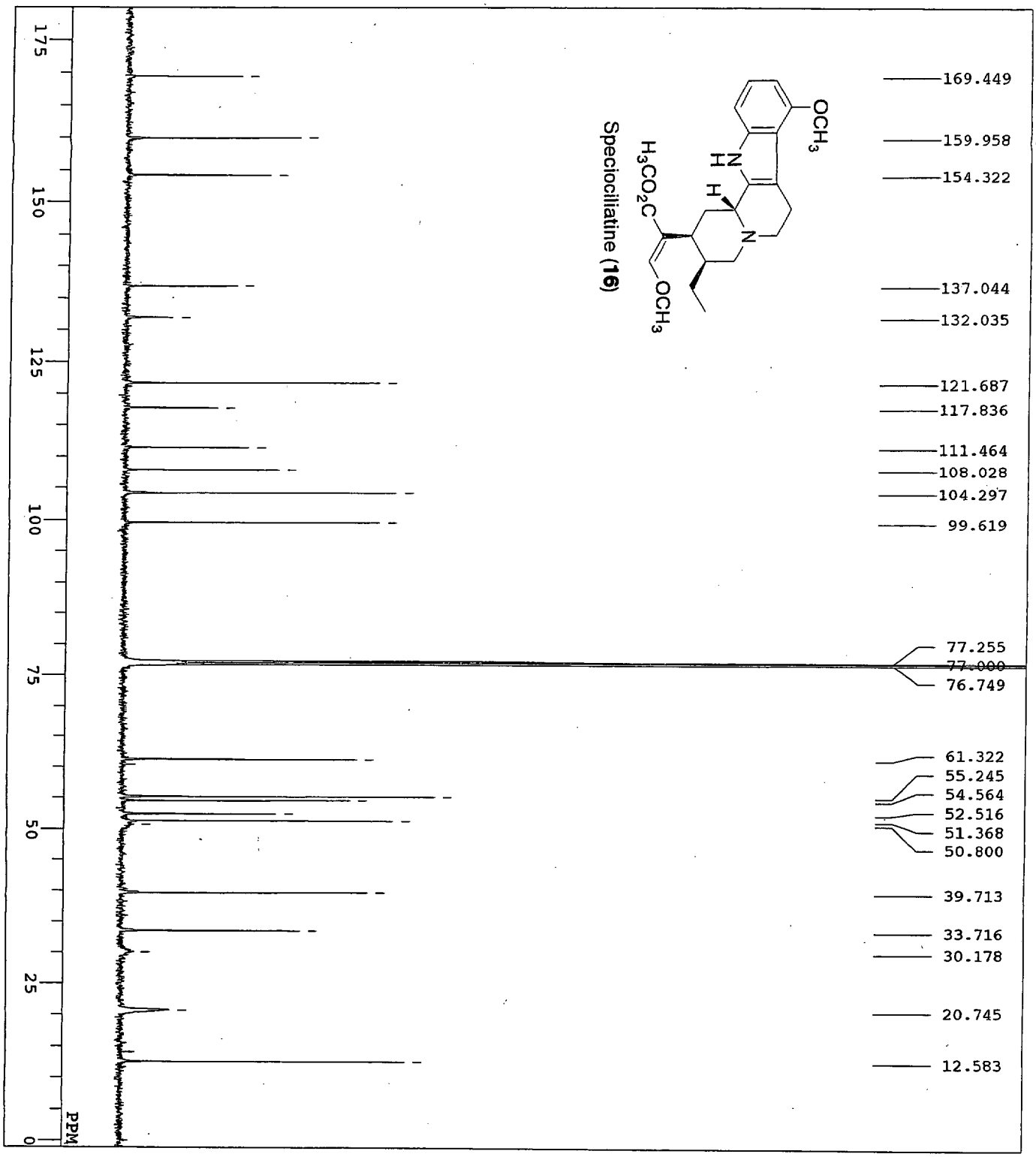
PW1 : 4.40 usec

OBNUC : 13C
OBFRQ : 125.65 MHz
OBSET : 127782.65 Hz

IRNUC : 1H
IRFRQ : 500.00 MHz
IRSET : 162410.00 Hz
IRATN : 511
IRRPW : 67.0 usec
IRBP1 : 30
IRBP2 : 6
IRANS : 0

ADBIT : 16
CTEMP : 21.2 c
CSPED : 12 Hz
SLVNT : CDCL3

RESOL : 0.46 Hz
BF : 2.00 Hz
T1 : 0.00 %
T2 : 0.00 %
T3 : 90.00 %
T4 : 100.00 %
REFVL : 77.00 ppm
XE : 21013.14 Hz
XS : 918.28 Hz
operator




```
DFILE  D:\NMR-data\Ishikawa\SKHI
COMNT  SPECIOCILIATINE-2-BCM
DATIM
OBNUC  13C
EXMOD  SINGL-BCM      LQNOR   Â@
OBFRQ      125.65 MHz
OBSET      120.00 KHz
OBFIN     7782.6 Hz
POINT       65536
FREQU    30030.0 Hz
SCANS        1000
ACQTM       2.182 sec
PD          1.500 sec
PW1            3.9 us
IRNUC  1H
CTEMP      21.5 c
SLVNT  CDCL3
EXREF      77.00 ppm
BF          3.00 Hz
RGAIN         23
```

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